INCH-POUND

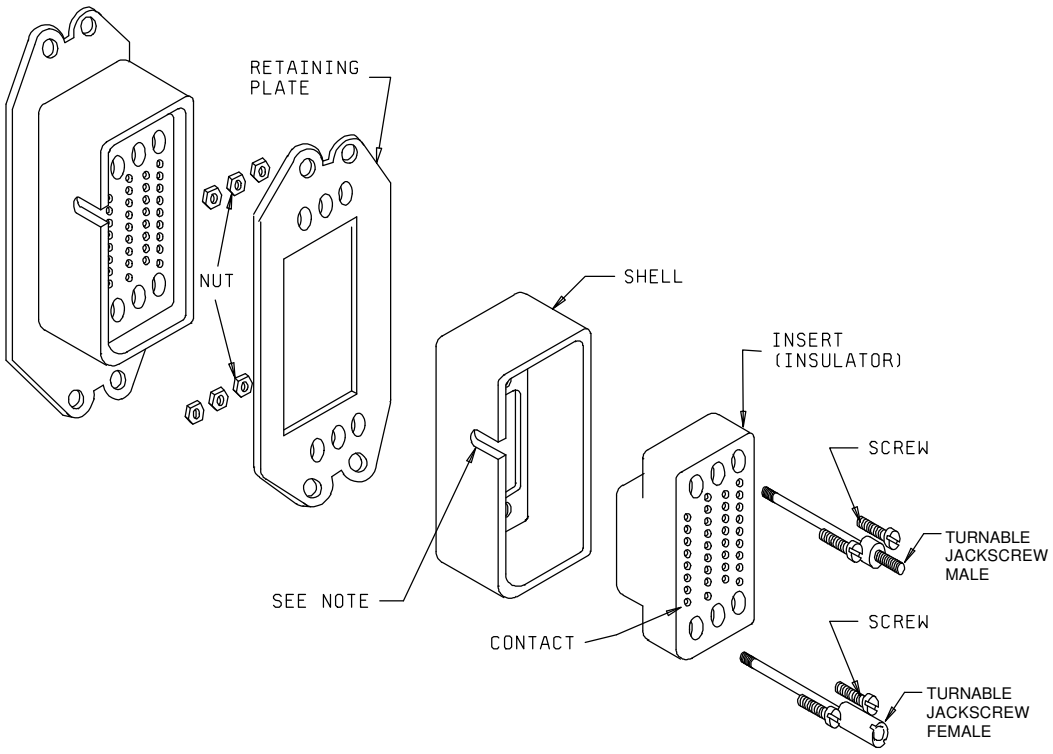
MIL-DTL-28748/4E
11 May 1999
SUPERSEDING
MIL-C-28748/4D
18 March 1985

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, CRIMP TYPE REMOVABLE SOCKET CONTACTS, SIZES 16 AND 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue MIL-DTL-28748 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.




Connector

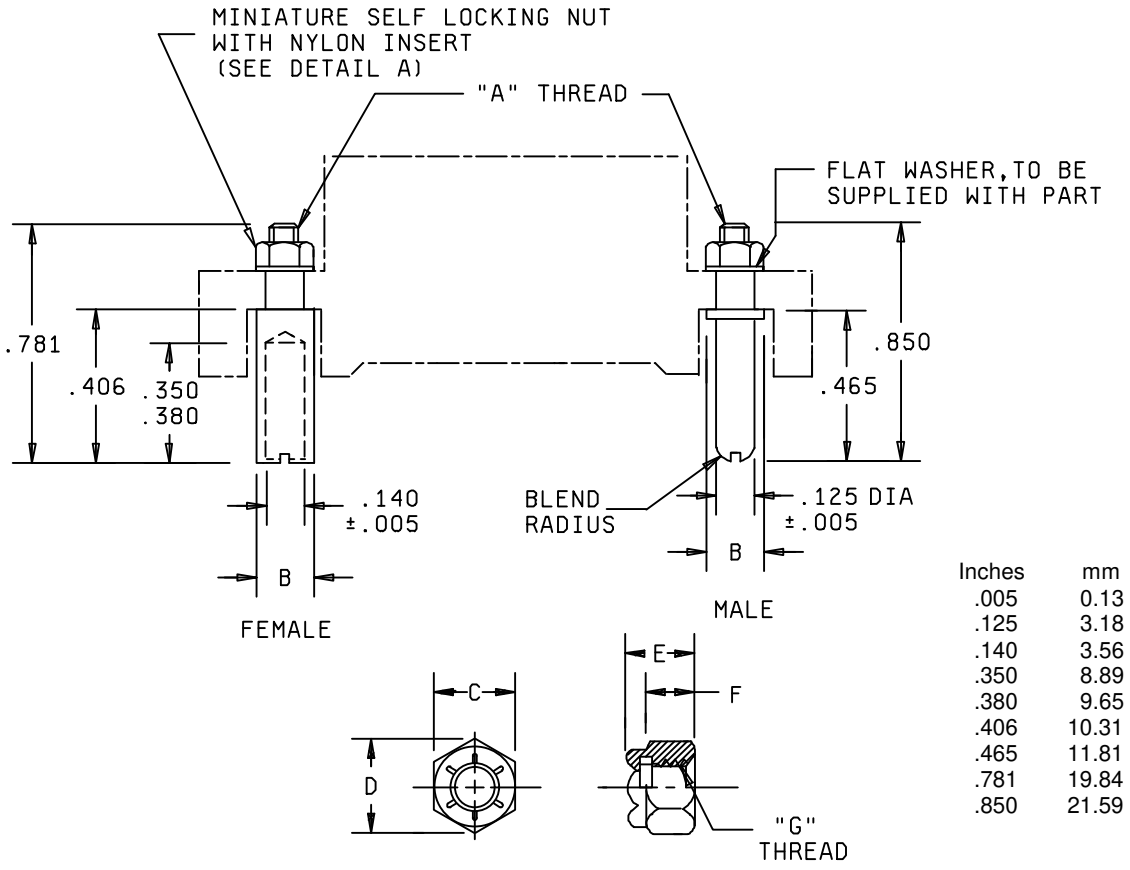
NOTE: The polarization feature (groove on the receptacle shell, pin on the plug shell) shall be adjacent to the side closest to the number 1 or A contact position.

Configuration A

FIGURE 1. Connector with shield, plug shell, jackscrews, and turnable jackscrews.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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| Inches | mm |
|---|---|
| .005 | 0.13 |
| .125 | 3.18 |
| .140 | 3.56 |
| .350 | 8.89 |
| .380 | 9.65 |
| .406 | 10.31 |
| .465 | 11.81 |
| .781 | 19.84 |
| .850 | 21.59 |

Detail A
Miniature self locking nut
(nylon insert)

Guidepin designator G

| Style | Connector size (number of contacts) | A Thread class 2 | B dia max | C min | C max | D ref | E min | E max | G Thread class 3 |
|---|---|---|---|---|---|---|---|---|---|
| Male | 9, 14, 20, 34, 42, 50, 66, 75 | 4-40 UNC-2A .112-40 UNC | .190 (4.83) | .181 (4.06) | .190 (4.83) | .205 (5.21) | .095 (2.41) | .125 (3.18) | 4-40 UNC-3B .112-40 UNC |
| Female | 9, 14, 20, 34, 42, 50, 66, 75 | 4-40 UNC-2A .112-40 UNC | .190 (4.83) | .181 (4.06) | .190 (4.83) | .205 (5.21) | .095 (2.41) | .125 (3.18) | 4-40 UNC-3B .112-40 UNC |
| Male | 104 | 6-32 UNC-2A .138-32 UNC | .190 (4.83) | .240 (6.10) | .252 (6.04) | .275 (6.99) | .133 (3.38) | .153 (3.89) | 6-32 UNC-3B .138-32 UNC |
| Female | 104 | 6-32 UNC-2A .138-32 UNC | .190 (4.83) | .240 (6.10) | .252 (6.04) | .275 (6.99) | .133 (3.38) | .153 (3.89) | 6-32 UNC-3B .138-32 UNC |

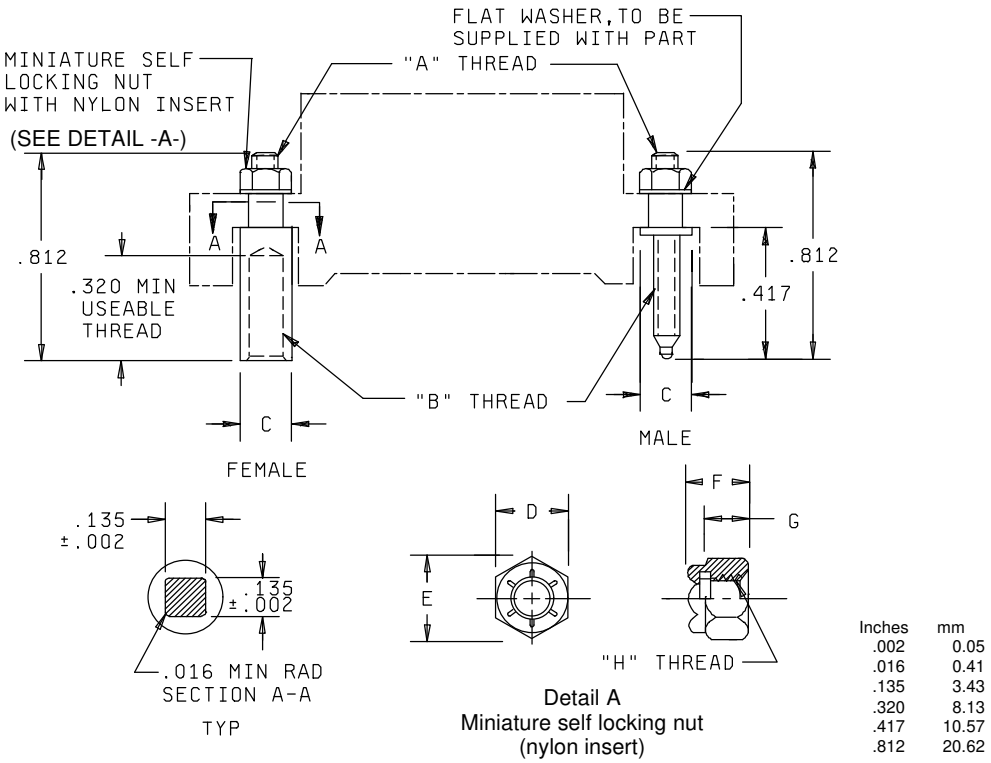
Connector with guidepins

Configuration B

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.



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Jackscrew designator F

| Style | Connector size (number of contacts) | A Thread class 2 | B Thread class 2 | C dia max |
|---|---|---|---|---|
| Male | 9, 14, 20, 34, 42, 50, 66, 75 | 4-40 UNC-2A .112-40 UNC | 6-32 UNC-2A .138-32 UNC | .190 (4.83) |
| Female | 9, 14, 20, 34, 42, 50, 66, 75 | 4-40 UNC-2A .112-40 UNC | 6-32 UNC-2B .138-32 UNC | .190 (4.83) |
| Male | 104 | 6-32 UNC-2A .138-32 UNC | 6-32 UNC-2A .138-32 UNC | .225 (5.72) |
| Female | 104 | 6-32 UNC-2A .138-32 UNC | 6-32 UNC-2B .138-32 UNC | .225 (5.72) |

Self locking nut

| Connector size (number of contacts) | D | | E ref | F | | G ref | H Thread |
|---|---|---|---|---|---|---|---|
| | min | max | | min | max | | |
| 9, 14, 20, 34, 42, 50, 66, 75 | .181 (4.06) | .190 (4.83) | .205 (5.21) | .095 (2.41) | .125 (3.18) | .070 (1.78) | 4-40 UNC-3B .112-40 UNC |
| 104 | .240 (6.10) | .252 (6.04) | .275 (6.99) | .133 (3.38) | .153 (3.89) | .081 (2.06) | 6-32 UNC-3B .138-32 UNC |

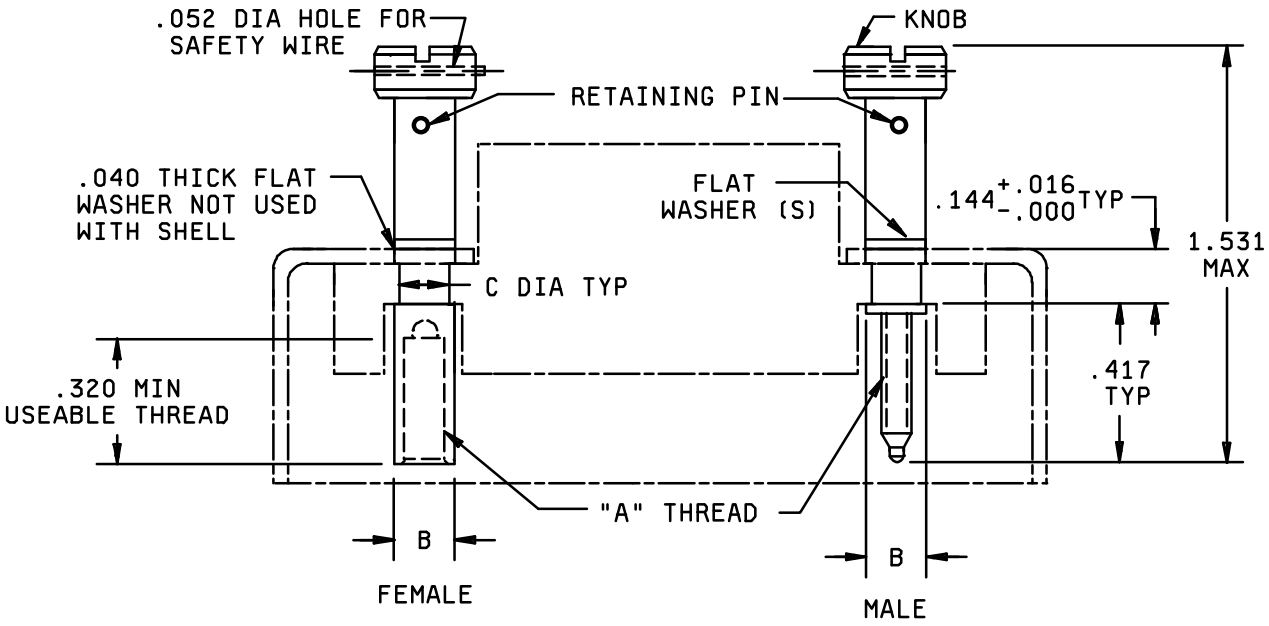
Connector with fixed jackscrews

Configuration C

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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| Inches | mm |
|---|---|
| .016 | 0.41 |
| .040 | 1.02 |
| .052 | 1.32 |
| .144 | 3.60 |
| .320 | 8.13 |
| .417 | 10.59 |
| 1.531 | 58.89 |

Jackscrew designator S

| Style | Connector size (number of contacts) | "A" Thread class 2 | "B" dia max | C DIA ±.003 (0.08 mm) |
|---|---|---|---|---|
| Male | 9, 14, 20, 34, 42, 50, 66, 75 | 6-32 UNC-2A .138-32 UNC | .190 (4.83) | .115 (2.92) |
| Female | 9, 14, 20, 34, 42, 50, 66, 75 | 6-32 UNC-2B .138-32 UNC | .190 (4.83) | .115 (2.92) |
| Male | 104 | 8-32 UNC-2A .164-32 UNC | .225 (5.72) | .137 (3.48)) |
| Female | 104 | 8-32 UNC-2B .164-32 UNC | .225 (5.72) | .137 (3.48) |

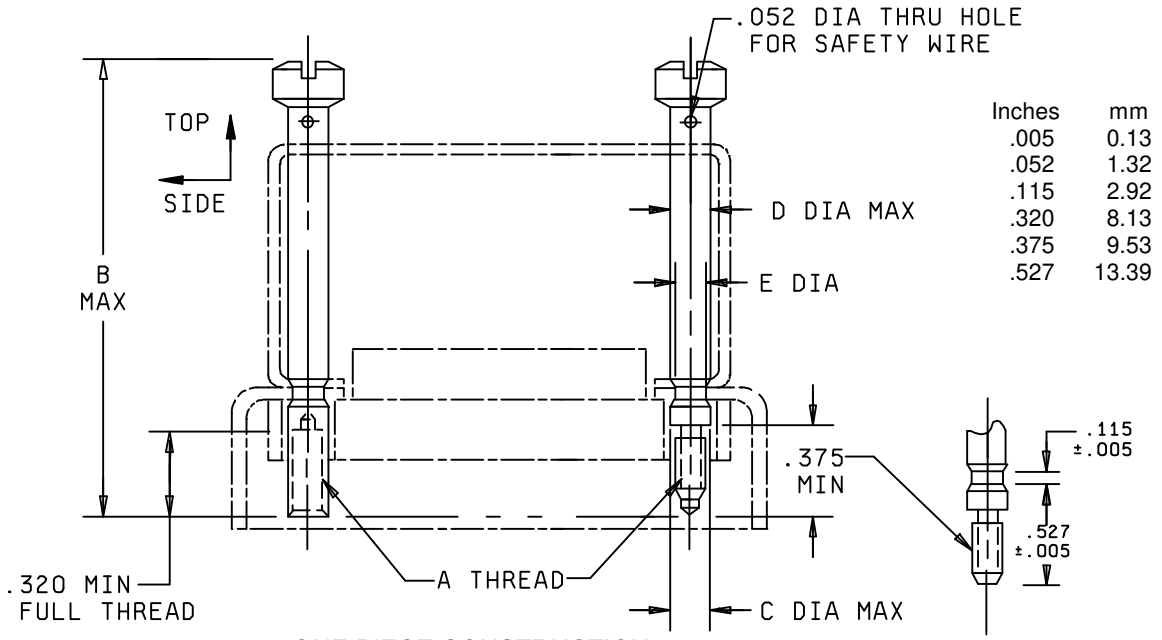
Connector with short turnable jackscrews

Configuration D

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.



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ONE PIECE CONSTRUCTION

Jackscrew designator L

| Style | Connector size (number of contacts) | "A" Thread class 2 | "B" max | C dia max | D dia max | E dia ±.002 (0.05 mm) |
|---|---|---|---|---|---|---|
| Male | 34, 42, 50, 66, 75 | 6-32 UNC<br>.138-32 UNC | 2.438<br>(61.93) | .192<br>(4.88) | .192<br>(4.88) | .111<br>(2.82) |
| Female | 34, 42, 50, 66, 75 | 6-32 UNC<br>.138-32 UNC | 2.438<br>(61.93) | .192<br>(4.88) | .192<br>(4.88) | .111<br>(2.82) |
| Male | 104 | 8-32 UNC<br>.164-32 UNC | 3.282<br>(83.36) | .225<br>(5.72) | .225<br>(5.72) | .140<br>(3.56) |
| Female | 104 | 8-32 UNC<br>.164-32 UNC | 3.282<br>(83.36) | .225<br>(5.72) | .225<br>(5.72) | .140<br>(3.56) |

For use with shields 3, 4, 9, A, D, and E or plug shells S thru Z, and receptacle shells 1 thru 8.

When plug shells S thru Z or receptacle polarizing shells 1 thru 8 are required to be used with a shield, use shield configurations 1, 2, B, and C.

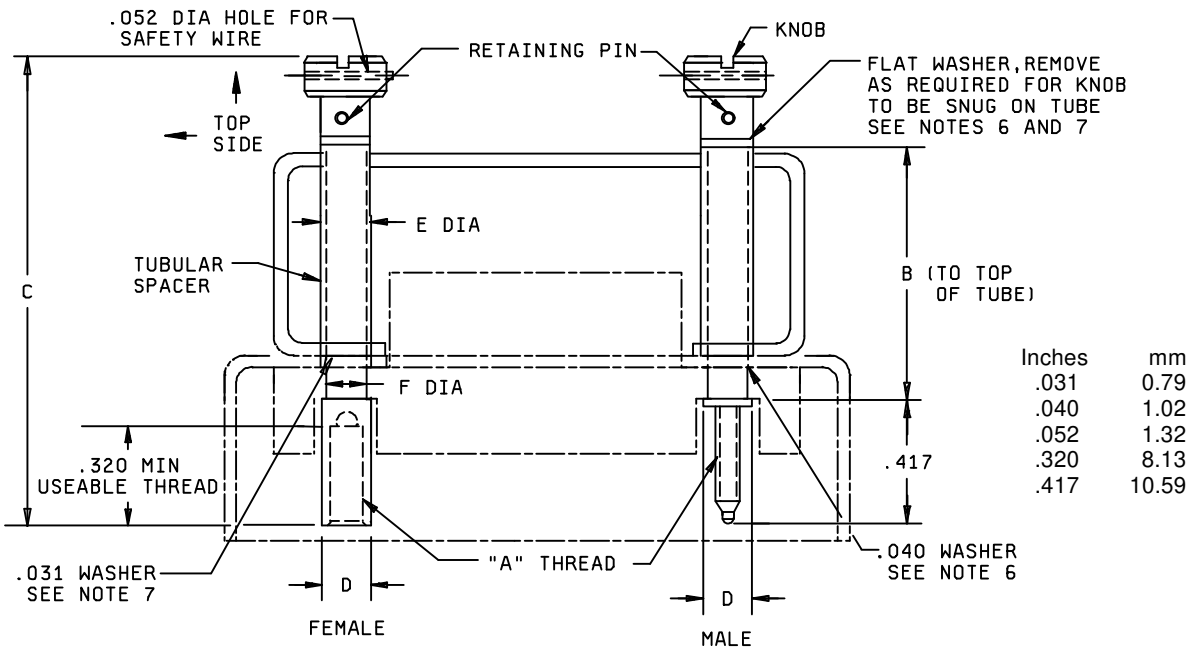
Connector with long turnable jackscrews

Configuration L

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.



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MULTIPIECE CONSTRUCTION

Jackscrew designator M

| Style | Connector size (number of contacts) | "A" Thread class 2 | "B" ±.016 (0.41 mm) | C max | D dia max | E dia ±.004 (0.10 mm) | F dia ±.003 |
|---|---|---|---|---|---|---|---|
| Male | 9, 14, 20, 34, 42, 50, 66, 75 | 6-32 UNC .138-32 UNC | 1.500 (43.10) | 2.467 (62.66) | .190 (4.83) | .188 (4.78) | .115 (2.92) |
| Female | 9, 14, 20, 34, 42, 50, 66, 75 | 6-32 UNC .138-32 UNC | 1.500 (43.10) | 2.467 (62.66) | .190 (4.83) | .188 (4.78) | .115 (2.92) |
| Male | 104 | 8-32 UNC .164-32 UNC | 2.495 (63.37) | 3.438 (87.33) | .225 (5.72) | .219 (5.56) | .137 (3.48) |
| Female | 104 | 8-32 UNC .164-32 UNC | 2.495 (63.37) | 3.438 (87.43) | .225 (5.72) | .219 (5.56) | .137 (3.48) |

When plug shells A through H and receptacle shells J through R are required to be used with a shield, use shield configurations 1, 2, B, and C.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with QQ-S-763 or QQ-S-764, classes 302, 303, 303se, 304 or 316 except for aluminum alloy knob and tubular, spacer, on multipiece construction.
5. Finish passivated in accordance with QQ-P-35 except for anodized knob on multipiece construction.
6. Washer .040 inch (1.02 mm) thick supplied with all sizes, not used with shell.
7. Washer .031 inch (0.79 mm) thick, supplied with shields 34 through 75.

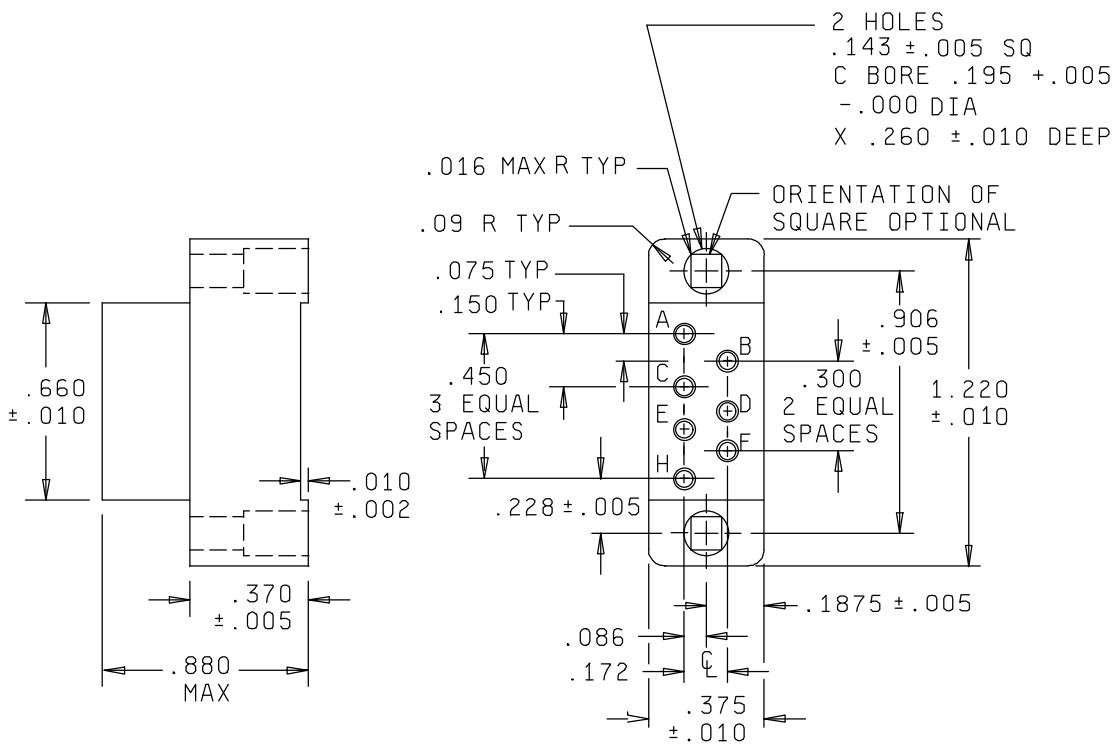
Connector with long turnable jackscrews

Configuration E

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .143 | 3.63 | .300 | 7.62 |
| .005 | 0.13 | .150 | 3.81 | .370 | 9.40 |
| .010 | 0.25 | .172 | 4.37 | .375 | 9.53 |
| .016 | 0.41 | .1875 | 4.76 | .450 | 11.43 |
| .075 | 1.91 | .195 | 4.95 | .660 | 16.76 |
| .086 | 2.18 | .228 | 5.79 | .880 | 22.35 |
| .09 | 2.29 | .260 | 6.60 | .906 | 23.01 |
| | | | | 1.220 | 30.99 |

The maximum insertion and withdrawal force (connector assembly) shall be 8.75 lbf (38.92 N) pounds.

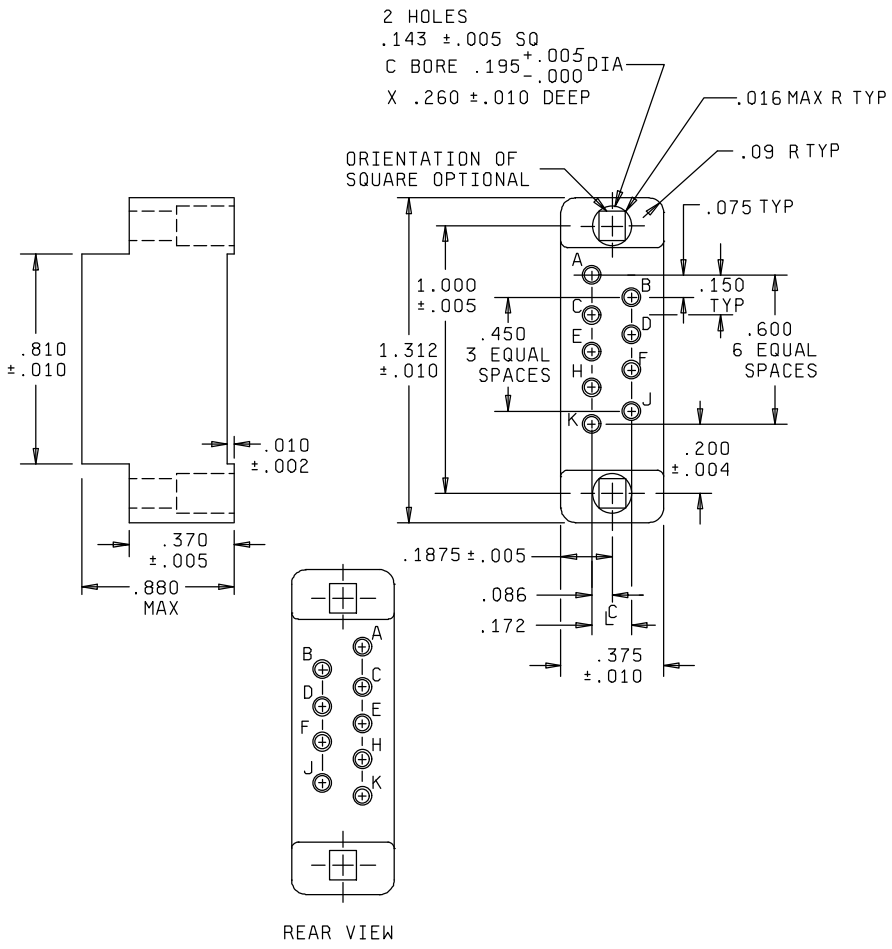
Insert with 7 contacts

Insert designators A and T

FIGURE 2. Insert arrangements.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .143 | 3.63 | .375 | 9.53 |
| .004 | 0.10 | .150 | 3.81 | .450 | 11.43 |
| .005 | 0.13 | .172 | 4.37 | .600 | 15.24 |
| .010 | 0.25 | .1875 | 4.76 | .810 | 20.57 |
| .016 | 0.41 | .195 | 4.95 | .856 | 21.74 |
| .075 | 1.95 | .200 | 5.08 | .880 | 22.35 |
| .086 | 2.00 | .260 | 6.60 | 1.000 | 25.40 |
| .09 | 2.60 | .370 | 9.40 | 1.312 | 33.32 |

The maximum insertion and withdrawal force (connector assembly) shall be 11.25 lbf (50 N) pounds.

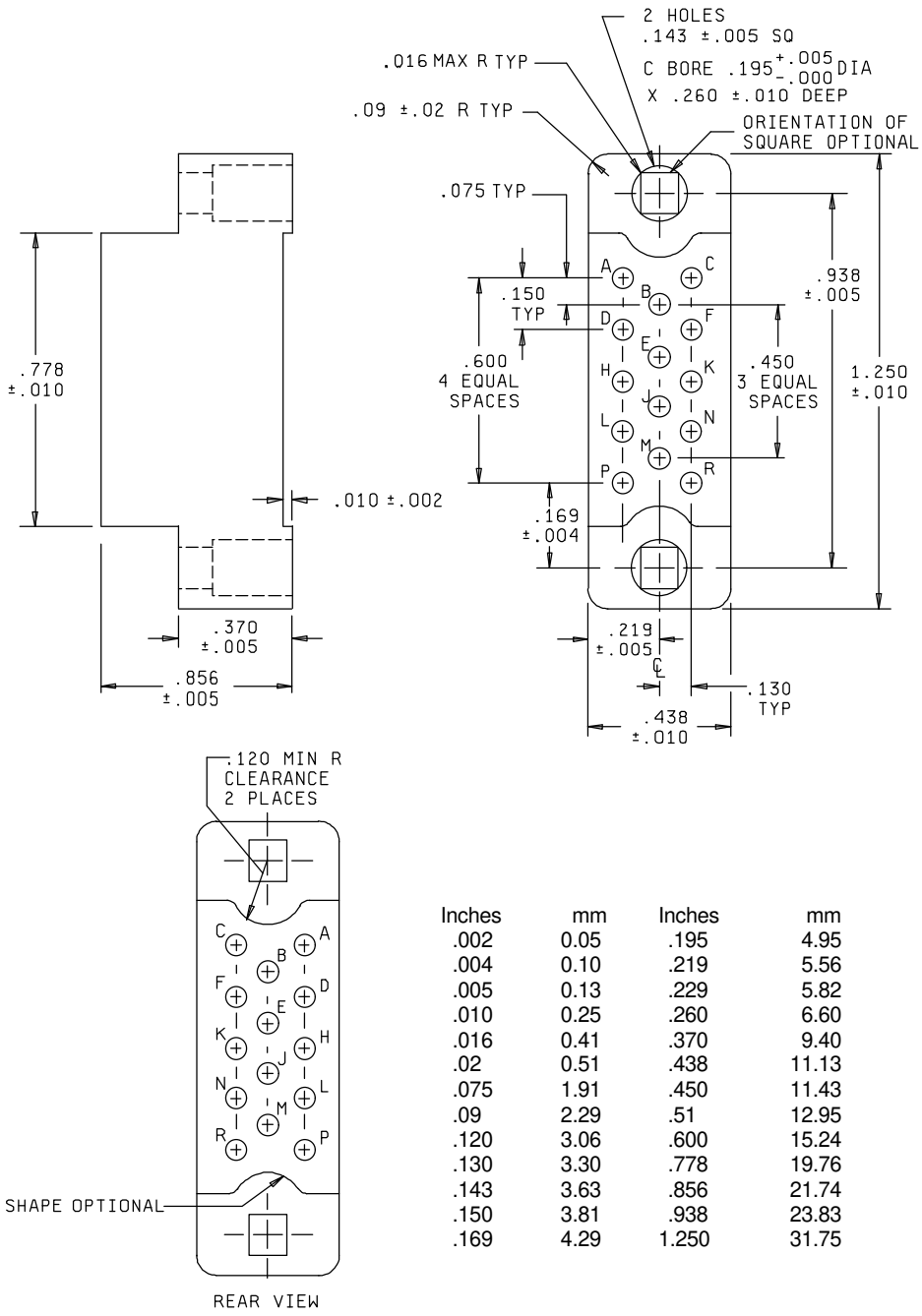
Insert with 9 contacts

Insert designators B and U

FIGURE 2. Insert arrangements.



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Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | 0.05 | .195 | 4.95 |
| .004 | 0.10 | .219 | 5.56 |
| .005 | 0.13 | .229 | 5.82 |
| .010 | 0.25 | .260 | 6.60 |
| .016 | 0.41 | .370 | 9.40 |
| .02 | 0.51 | .438 | 11.13 |
| .075 | 1.91 | .450 | 11.43 |
| .09 | 2.29 | .51 | 12.95 |
| .120 | 3.06 | .600 | 15.24 |
| .130 | 3.30 | .778 | 19.76 |
| .143 | 3.63 | .856 | 21.74 |
| .150 | 3.81 | .938 | 23.83 |
| .169 | 4.29 | 1.250 | 31.75 |

The maximum insertion and withdrawal force (connector assembly) shall be 17.5 lbf (77.84 N).

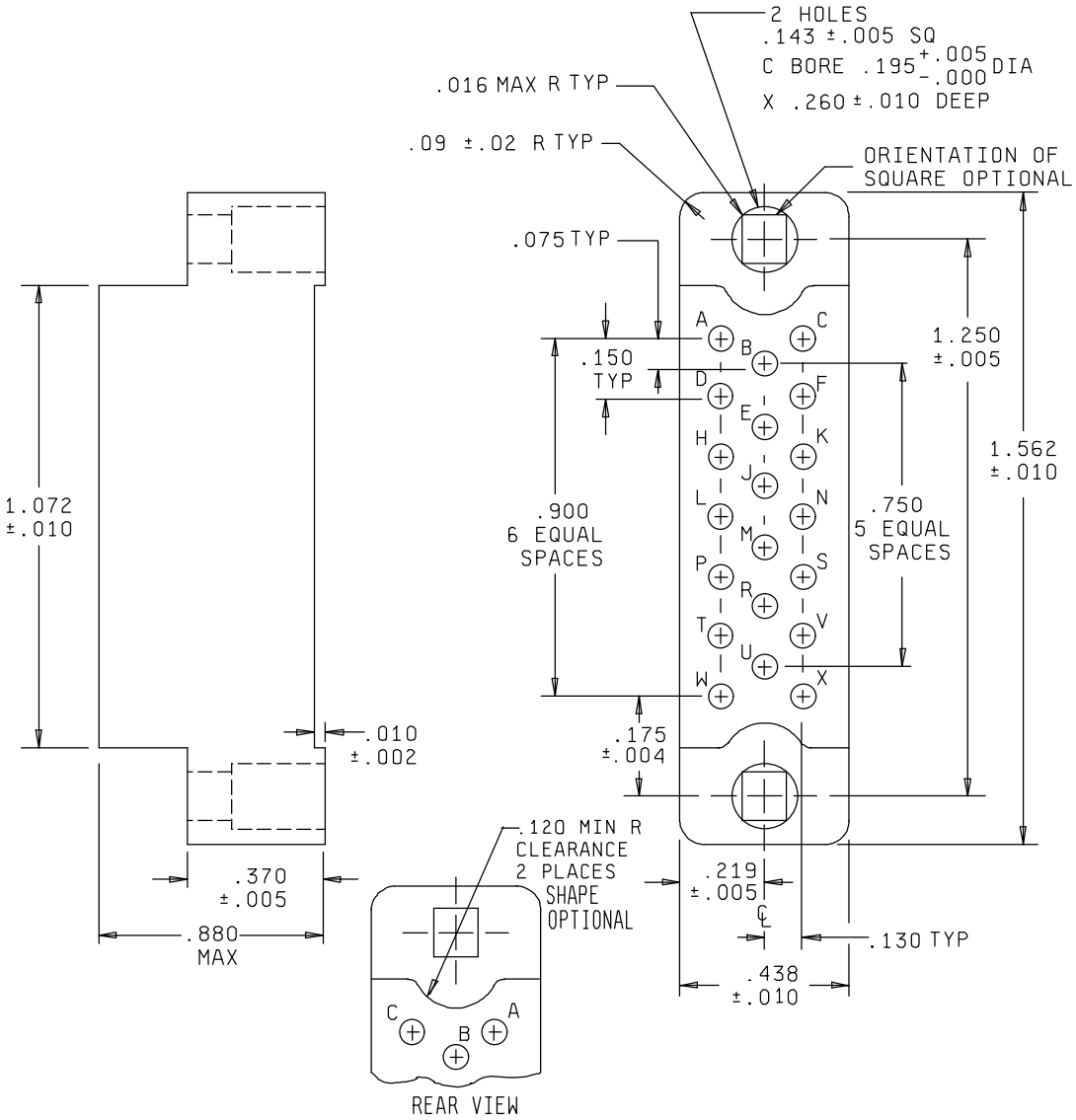
Insert with 14 contacts

Insert designators C and V

FIGURE 2. Insert arrangements - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .120 | 3.06 | .438 | 11.13 |
| .004 | 0.10 | .130 | 3.30 | .51 | 12.95 |
| .005 | 0.13 | .143 | 3.63 | .750 | 19.05 |
| .010 | 0.25 | .150 | 3.81 | .880 | 22.35 |
| .016 | 0.41 | .175 | 4.45 | .900 | 22.86 |
| .02 | 0.51 | .195 | 4.95 | 1.072 | 27.23 |
| .075 | 1.91 | .219 | 5.56 | 1.250 | 31.75 |
| .09 | 2.29 | .260 | 6.60 | 1.562 | 39.67 |
| | | .370 | 9.40 | 2.290 | 58.17 |

The maximum insertion and withdrawal force (connector assembly) shall be 25 lbf (111.21 N).

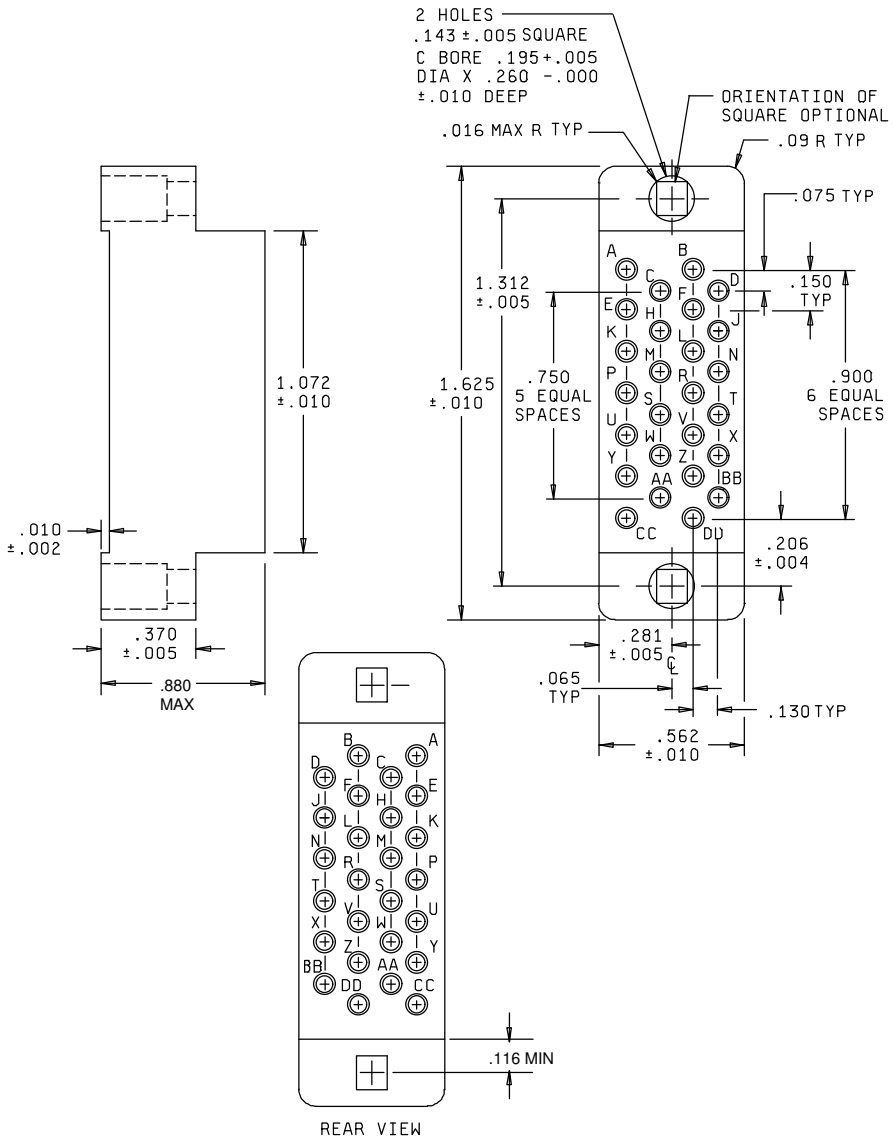
Insert with 20 contacts

Insert designators D and W

FIGURE 2. Insert arrangements - Continued.



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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .004 | 0.10 |
| .005 | 0.13 |
| .010 | 0.25 |
| .016 | 0.41 |
| .065 | 1.65 |
| .075 | 1.91 |
| .09 | 2.29 |
| .116 | 2.95 |
| .130 | 3.30 |
| .143 | 3.63 |
| .150 | 3.81 |
| .195 | 4.95 |
| .206 | 5.23 |
| .260 | 6.60 |
| .281 | 7.14 |
| .370 | 9.40 |
| .562 | 14.27 |
| .750 | 19.05 |
| .856 | 21.74 |
| .880 | 22.35 |
| .900 | 22.86 |
| 1.072 | 27.23 |
| 1.312 | 33.32 |
| 1.625 | 41.28 |

The maximum insertion and withdrawal force (connector assembly) shall be 32.5 lbf (144.57 N).

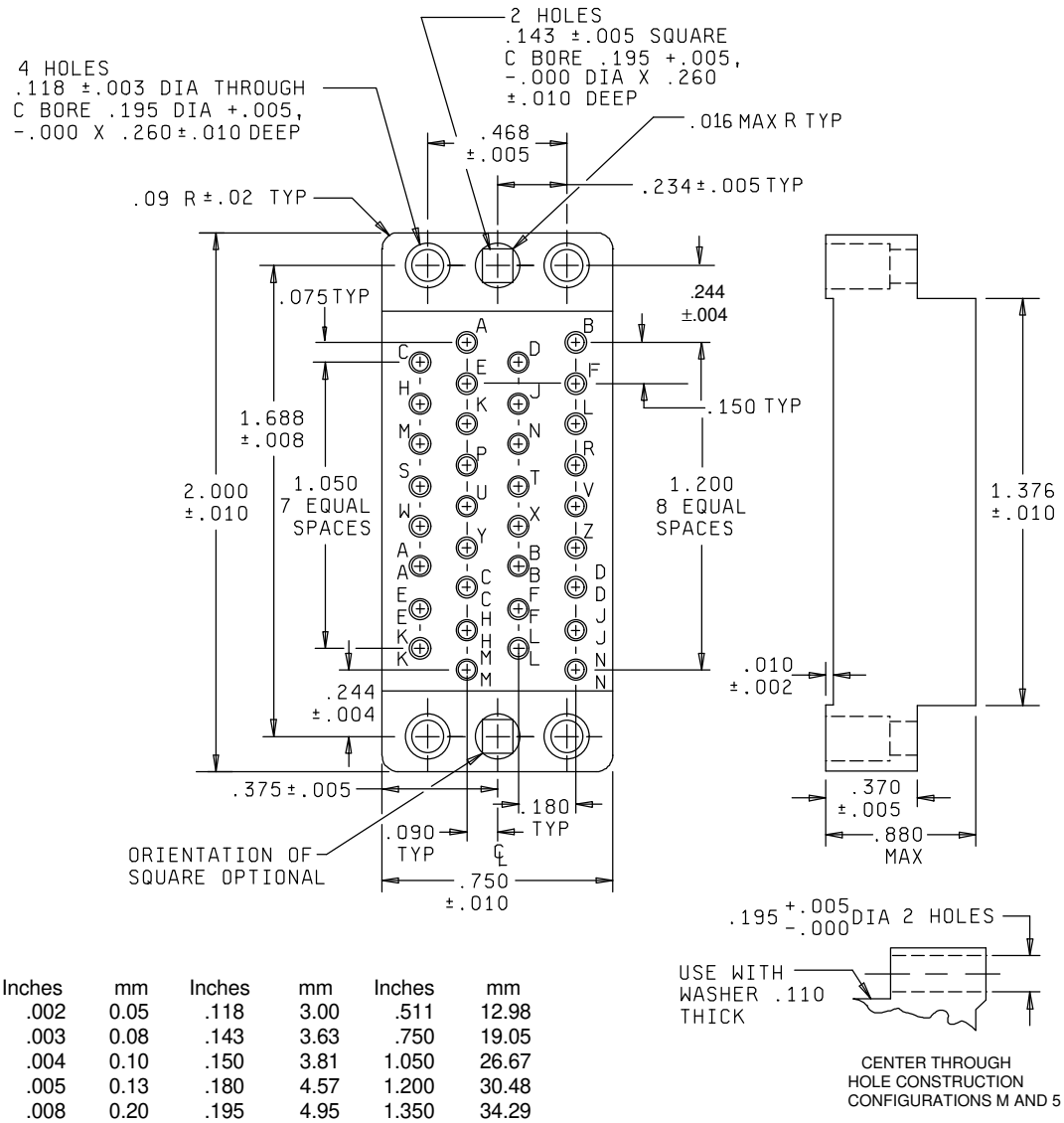
Insert with 26 contacts

Insert designators E and X

FIGURE 2. Insert arrangements - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .118 | 3.00 | .511 | 12.98 |
| .003 | 0.08 | .143 | 3.63 | .750 | 19.05 |
| .004 | 0.10 | .150 | 3.81 | 1.050 | 26.67 |
| .005 | 0.13 | .180 | 4.57 | 1.200 | 30.48 |
| .008 | 0.20 | .195 | 4.95 | 1.350 | 34.29 |
| .09 | 2.29 | .234 | 5.94 | 1.376 | 34.95 |
| .010 | 0.25 | .244 | 6.20 | 1.500 | 38.10 |
| .016 | 0.41 | .250 | 6.35 | 1.672 | 42.47 |
| .075 | 1.91 | .260 | 6.60 | 1.688 | 42.88 |
| .090 | 2.29 | .370 | 9.40 | 2.000 | 50.80 |
| .110 | 2.79 | .375 | 9.53 | 2.290 | 58.17 |
| | | .468 | 11.68 | 2.312 | 58.72 |

The maximum insertion and withdrawal force (connector assembly) shall be 42.5 lbf (189.05 N).

Configurations F and Y - Suitable for all applications, except for one-piece jackscrews.
Configurations M and 5 - For one-piece jackscrews (through hole construction).

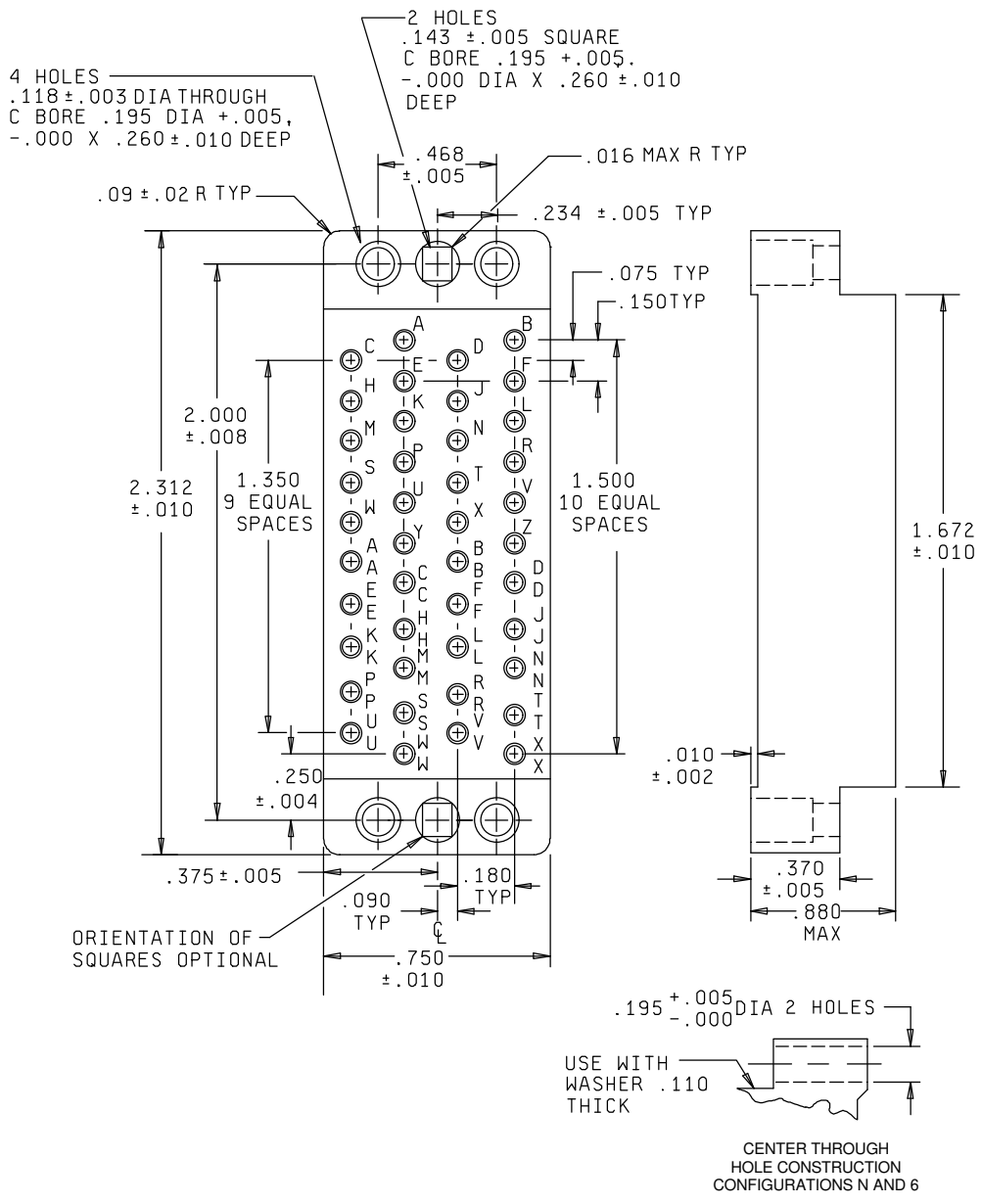
Insert with 34 contacts

Insert designators F, M, Y, and 5

FIGURE 2. Insert arrangements – Continued.



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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .02 | 0.51 |
| .075 | 1.91 |
| .09 | 2.3 |
| .090 | 2.29 |
| .110 | 2.79 |
| .118 | 3.00 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .250 | 6.35 |
| .260 | 6.60 |
| .370 | 9.40 |
| .375 | 9.53 |
| .468 | 11.13 |
| .51 | 12.95 |
| .750 | 19.05 |
| .856 | 21.74 |
| .880 | 22.35 |
| 1.350 | 34.29 |
| 1.500 | 38.10 |
| 1.672 | 42.47 |
| 2.000 | 50.80 |
| 2.290 | 58.17 |
| 2.312 | 58.72 |

The maximum insertion and withdrawal force (connector assembly) shall be 52.5 lbf (233.53 N).

Configurations G and Z - Suitable for all applications, except for one-piece jackscrews.
Configurations N and 6 - For one-piece jackscrews (through hole construction).

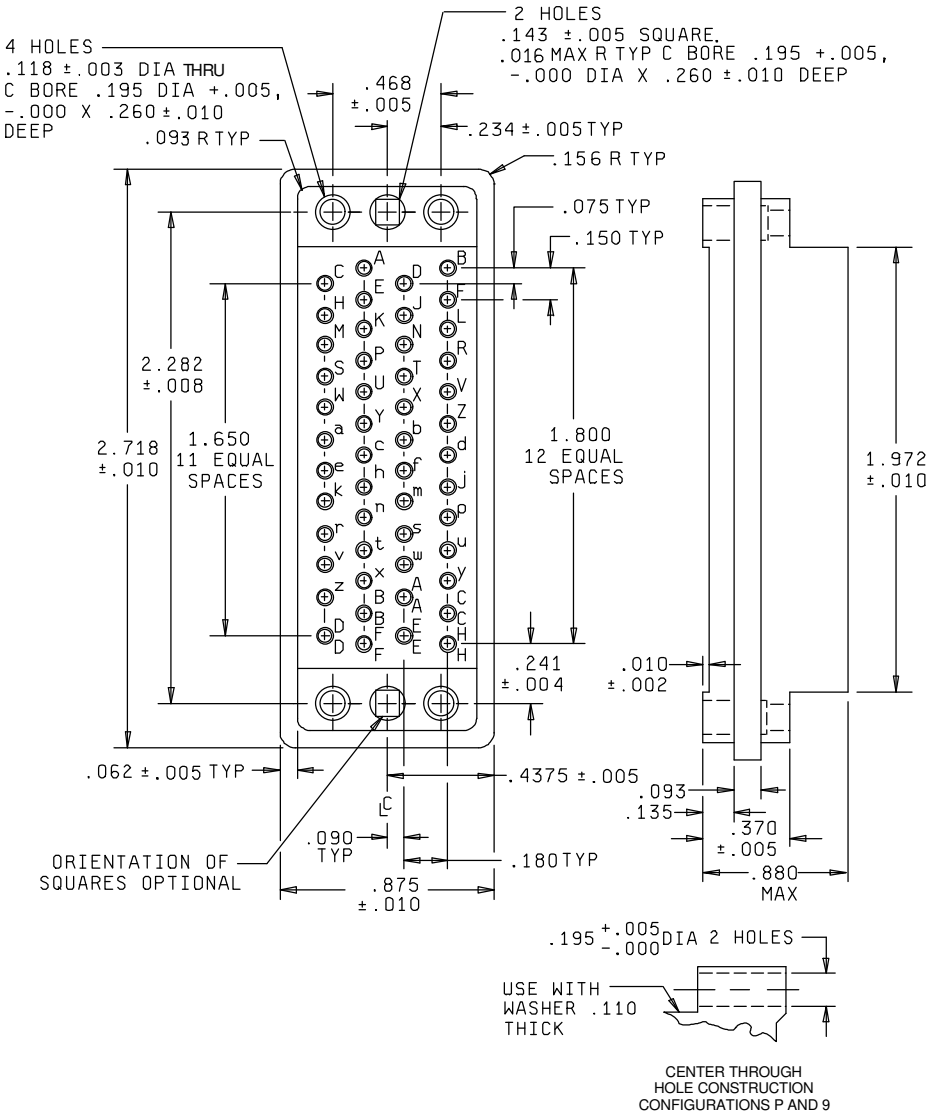
Insert with 42 contacts

Insert designators G, N, Z, and 6

FIGURE 2. Insert arrangements - Continued.



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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .062 | 1.57 |
| .075 | 1.91 |
| .090 | 2.29 |
| .093 | 2.36 |
| .110 | 2.79 |
| .118 | 3.00 |
| .135 | 3.43 |
| .143 | 3.63 |
| .150 | 3.81 |
| .156 | 3.96 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .241 | 6.12 |
| .260 | 6.60 |
| .370 | 9.40 |
| .4375 | 11.11 |
| .468 | 11.13 |
| .856 | 21.74 |
| .875 | 22.23 |
| .880 | 22.35 |
| 1.650 | 41.91 |
| 1.800 | 45.72 |
| 1.972 | 50.09 |
| 2.000 | 50.80 |
| 2.282 | 57.96 |
| 2.718 | 69.04 |

The maximum insertion and withdrawal force (connector assembly) shall be 62.5 lbf (278.01 N).

Configurations H and 1 - Suitable for all applications, except one-piece jackscrews.
Configurations P and 7 - For one-piece jackscrews and hood only (through hole construction).

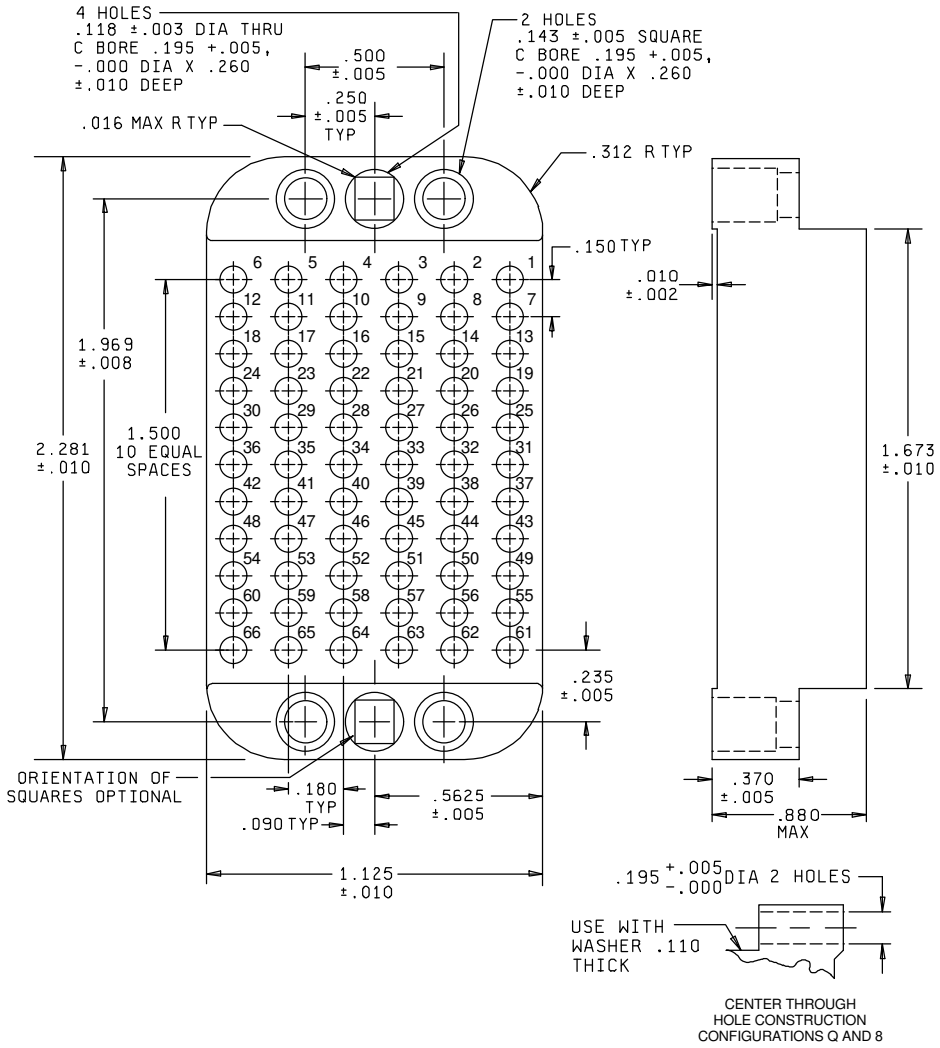
Insert with 50 contacts

Insert designators H, P, 1, and 7

FIGURE 2. Insert arrangements - Continued.



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Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .090 | 2.29 |
| .110 | 2.79 |
| .118 | 3.00 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .195 | 4.95 |
| .235 | 5.97 |
| .250 | 6.35 |
| .260 | 6.60 |
| .312 | 7.92 |
| .370 | 9.40 |
| .500 | 12.70 |
| .5625 | 14.288 |
| .880 | 22.35 |
| 1.125 | 28.58 |
| 1.500 | 38.10 |
| 1.673 | 42.49 |
| 1.969 | 50.01 |
| 2.281 | 57.94 |

The maximum insertion and withdrawal force (connector assembly) shall be 82.5 lbf (366.98 N).

Configurations J and 2 - Suitable for all applications, except one-piece jackscrews.
Configurations Q and 8 - For one-piece jackscrews (through hole construction).

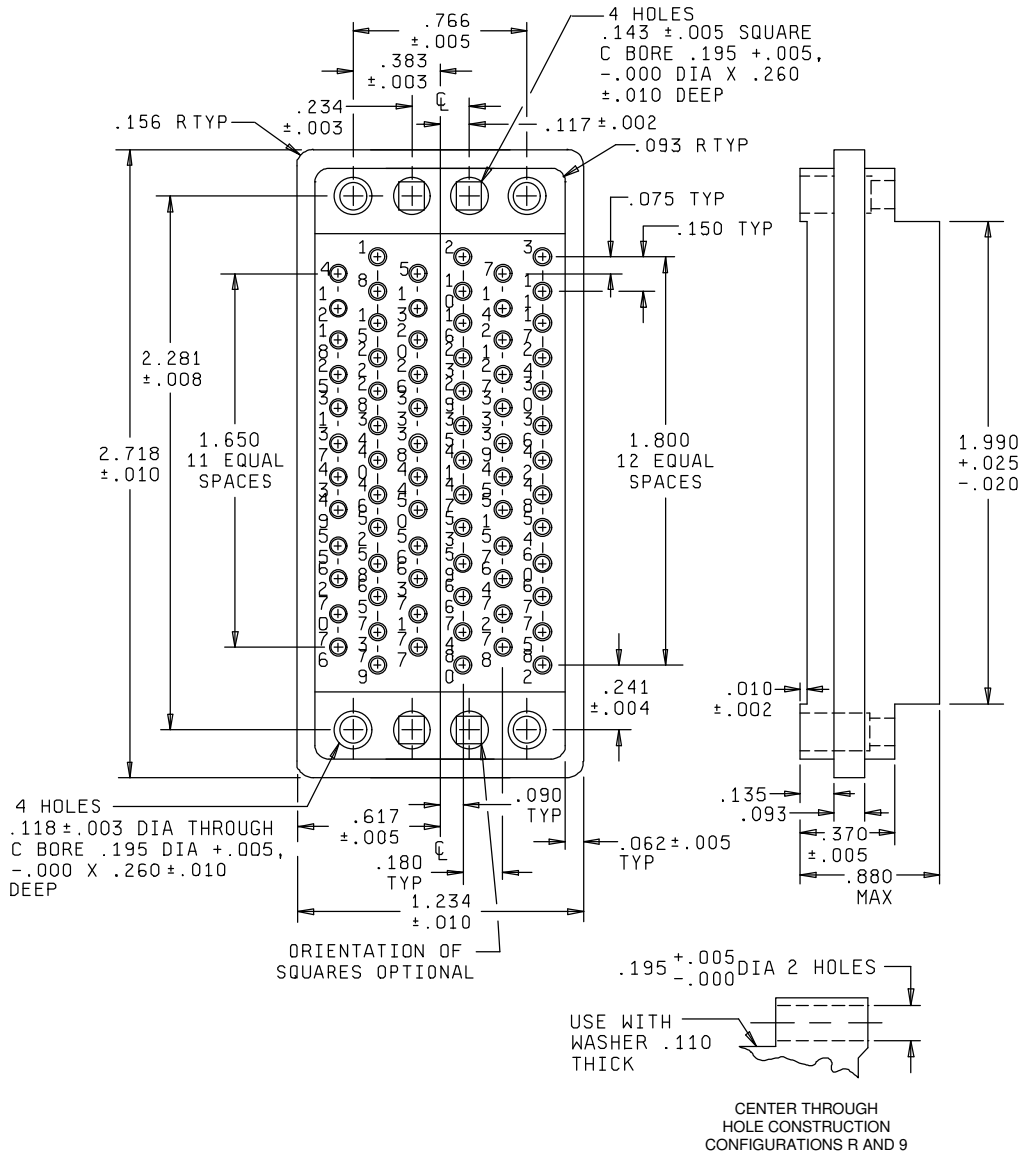
Insert with 66 contacts

Insert designators J, Q, 2 and 8

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




CENTER THROUGH HOLE CONSTRUCTION CONFIGURATIONS R AND 9

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.41 |
| .020 | 0.51 |
| .025 | 0.64 |
| .062 | 1.51 |
| .075 | 1.91 |
| .090 | 2.29 |
| .093 | 2.36 |
| .110 | 2.79 |
| .117 | 2.97 |
| .118 | 3.00 |
| .135 | 3.43 |
| .143 | 3.63 |
| .150 | 3.81 |
| .156 | 3.96 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .241 | 6.12 |
| .260 | 6.60 |
| .370 | 9.40 |
| .383 | 9.73 |
| .617 | 15.67 |
| .766 | 19.46 |
| .880 | 22.35 |
| 1.234 | 31.34 |
| 1.650 | 41.91 |
| 1.800 | 45.72 |
| 1.990 | 50.55 |
| 2.281 | 57.94 |
| 2.718 | 69.04 |

The maximum insertion and withdrawal force (connector assembly) shall be 93.7 lbf (416.80 N).

Configurations K and 3 - Suitable for all applications, except one-piece jackscrews.
Configurations R and 9 - For one-piece jackscrews (through hole construction).

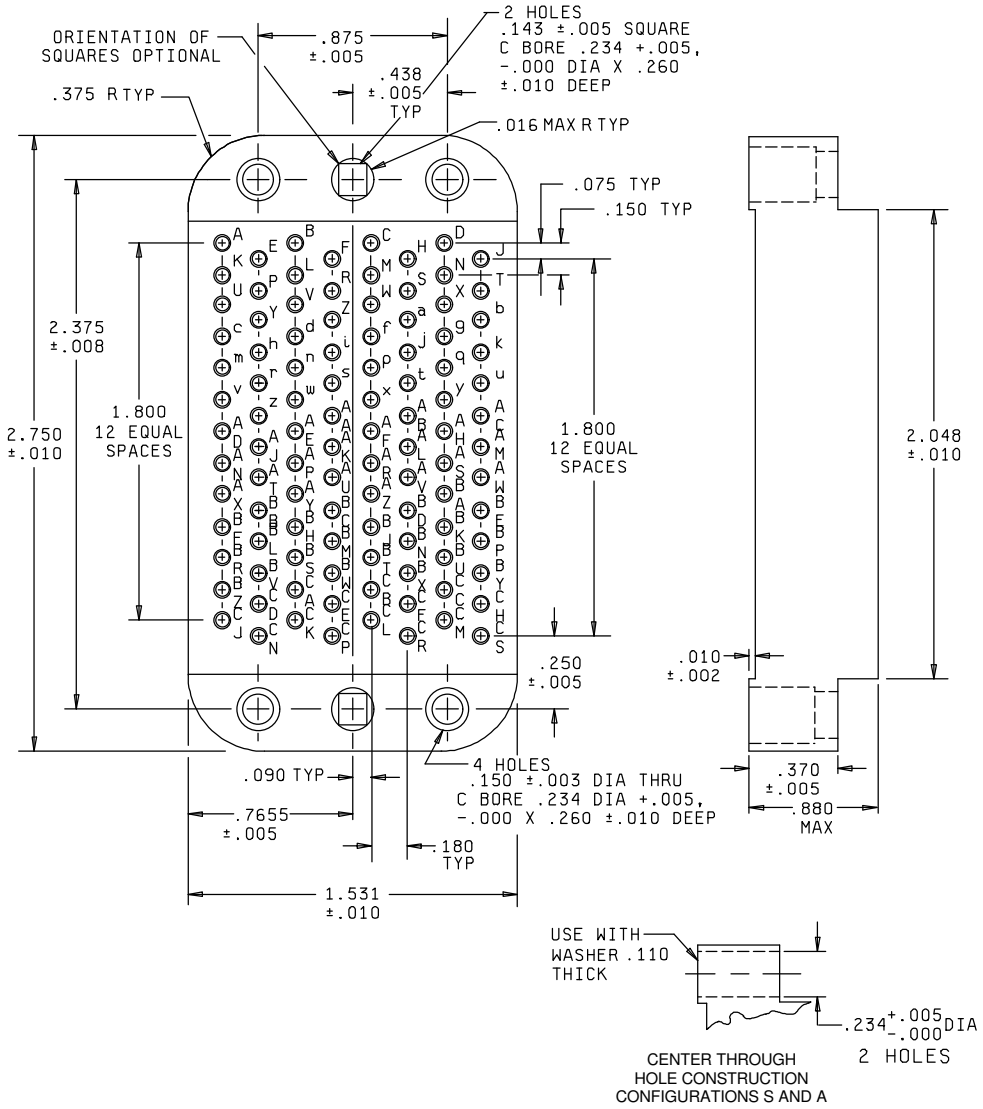
Insert with 75 contacts

Insert designators K, R, 3, and 9

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .075 | 1.91 |
| .090 | 2.79 |
| .110 | 2.79 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .234 | 5.94 |
| .250 | 6.35 |
| .260 | 6.60 |
| .370 | 9.40 |
| .375 | 9.53 |
| .438 | 11.13 |
| .7655 | 19.444 |
| .875 | 22.23 |
| .880 | 22.35 |
| 1.531 | 38.89 |
| 1.800 | 45.72 |
| 2.048 | 52.02 |
| 2.375 | 60.33 |
| 2.750 | 69.85 |

The maximum insertion and withdrawal force (connector assembly) shall be 130 lbf (578.27 N).

Configuration L - Suitable for all applications.
Configurations S - For long jackscrews and hood only (through hole alternate construction).

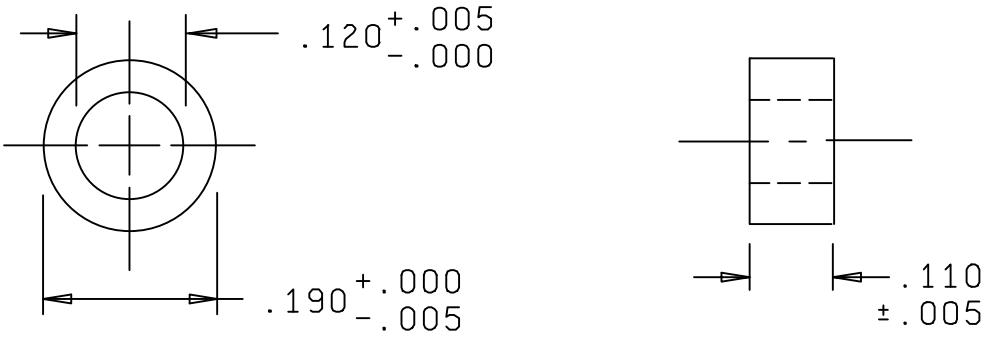
Insert with 104 contacts

Insert designators L, S, 4, and A

FIGURE 2. Insert arrangements - Continued.

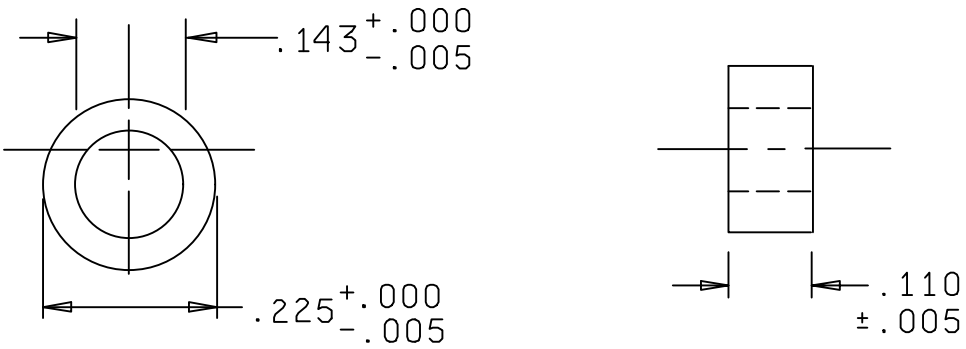


Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .110 | 2.79 |
| .120 | 3.05 |
| .143 | 3.63 |
| .190 | 4.83 |
| .225 | 5.72 |

For use With inserts M, N, P, Q, R (34, 42, 50, 66, AND 75).



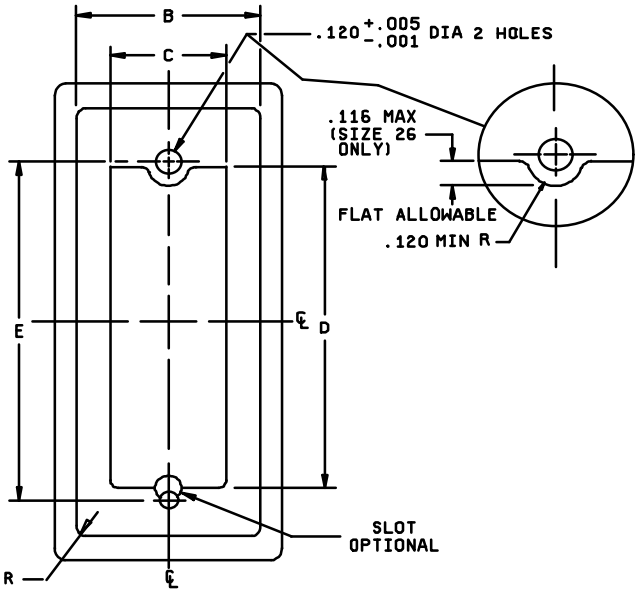

For use with insert S (104 contacts).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals. and ±.015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ±.004 inch (0.10 mm) . Tolerance between any two contact centers, other than adjacent contacts, shall be ±.006 inch (0.15 mm).
5. Contact hole details shall be in accordance with MIL-DTL-28748.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Designed for use with MIL-DTL-28748 size 16 and M39029/43-271 size 20 contacts.
8. Insert material shall be in accordance with MIL-DTL-28748.
9. Washer and self locking nut: Corrosion resistant steel, in accordance with QQ-S-763 or QQ-S-764, classes 302, 303, 303Se, 304, or 316.
10. Washer finish: Passivated in accordance with QQ-P-35 and self locking nut.

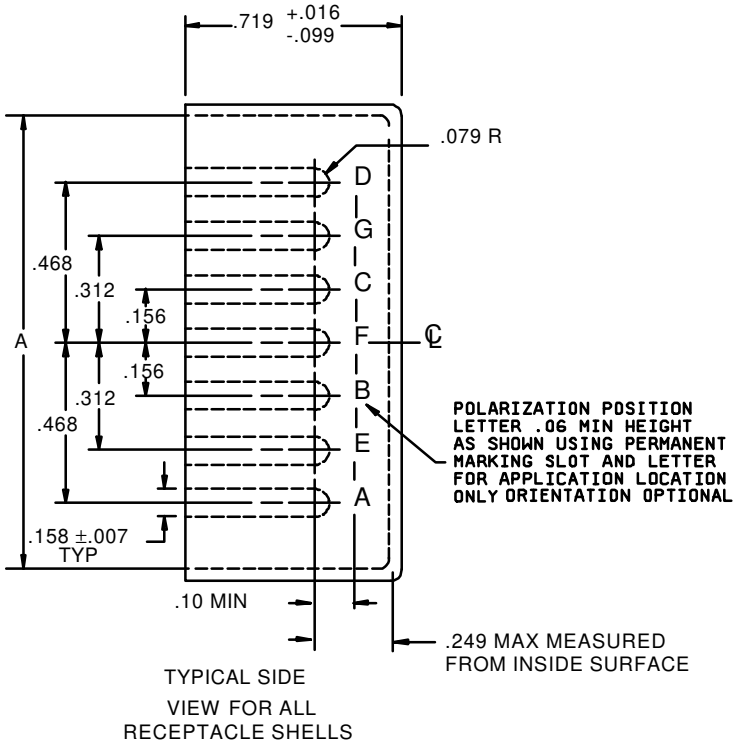
FIGURE 2. Insert arrangements - Continued.



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Check the source to verify that this is the current version before use.




STYLE FOR 9, 14, 20 AND 26 CONTACT CONNECTORS (J THRU R)



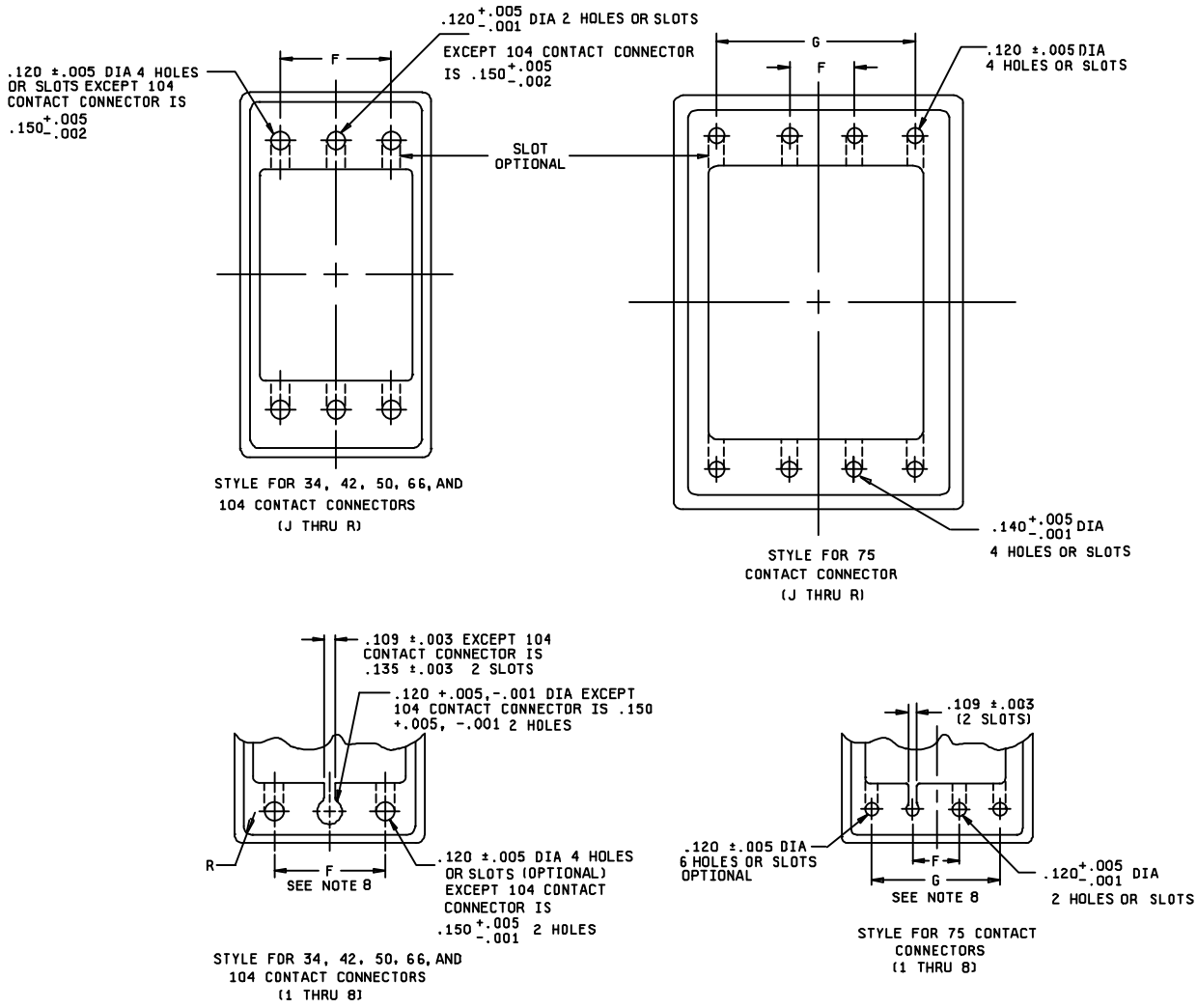

TYPICAL SIDE VIEW FOR ALL RECEPTACLE SHELLS

Receptacle shell polarizing

FIGURE 3. Shell configurations



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .125 | 3.18 |
| .002 | 0.05 | .135 | 3.43 |
| .003 | 0.08 | .140 | 3.56 |
| .005 | 0.13 | .150 | 3.81 |
| .007 | 0.18 | .156 | 3.96 |
| .079 | 2.01 | .158 | 4.01 |
| .099 | 2.51 | .234 | 5.94 |
| .10 | 2.54 | .312 | 7.92 |
| .109 | 2.77 | .452 | 11.48 |
| .116 | 2.95 | .468 | 11.89 |
| .120 | 3.05 | .719 | 18.26 |
| | | .936 | 23.77 |

Receptacle shell polarizing

FIGURE 3. Shell configurations - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

| Polarization designators | Connector size (number of contacts) | A min | B min | C min | D min | E ±.005 (0.13 mm) | F ±.005 (0.13 mm) | G ±.005 (0.13 mm) | R |
|---|---|---|---|---|---|---|---|---|---|
| J thru R | 9 | 1.448 (36.78) | .512 (13.00) | .375 (9.52) | .864 (21.94) | 1.000 (25.40) | --- | | .141 (3.58) |
| J thru R | 14 | 1.388 (35.36) | .575 (14.60) | .440 (11.18) | .800 (20.32) | .938 (23.82) | --- | | .141 |
| J thru R | 20 | 1.702 (43.23) | .575 (14.60) | .440 (11.18) | 1.097 (27.86) | 1.250 (31.75) | --- | | .141 |
| J thru R | 26 | 1.763 (44.78) | .696 (17.68) | .565 (14.35) | 1.182 (30.02) | 1.312 (33.32) | --- | | .141 (3.58) |
| J thru R | 34 | 2.138 (54.30) | .888 (22.56) | .745 (18.92) | 1.400 (35.76) | 1.688 (42.88) | .468 (11.89) | | .141 (3.58) |
| J thru R | 42 | 2.450 (62.23) | .888 (22.56) | .745 (18.92) | 1.700 (43.18) | 2.000 (50.80) | .468 (11.89) | | .141 (3.58) |
| J thru R | 50 | 2.856 (72.54) | 1.015 (25.78) | .745 (18.92) | 2.026 (51.46) | 2.281 (57.94) | .468 (11.89) | | .188 4/ (4.78) |
| J thru R | 66 | 2.420 (61.47) | 1.264 (32.11) | 1.115 (28.32) | 1.696 (43.08) | 1.969 (50.01 | .500 (12.70) | | .375 (9.52) |
| J thru R | 75 | 2.855 (72.52) | 1.370 (34.80) | 1.104 (28.04) | 2.026 (51.46) | 2.281 (57.94) | .234 (5.94) | .766 (19.46) | .203 (5.16) |
| J thru R | 104 | 2.900 (73.66) | 1.680 (42.67) | 1.476 (37.49) | 2.073 (52.65) | 2.375 (60.32) | .875 (22.22) | | .422 (10.72) |
| 1 thru 8 (see note 8) | 34 | 2.138 (54.30) | .888 (22.56) | .745 (18.92) | 1.400 (35.56) | 1.688 (42.88) | .468 (11.89) | | .141 (3.58) |
| 1 thru 8 (see note 8) | 42 | 2.450 (62.23) | .888 (22.56) | .745 (18.92) | 1.700 (43.18) | 2.000 (50.80) | .468 (11.89) | | .141 (3.58) |
| 1 thru 8 (see note 8) | 50 | 2.856 (72.54) | 1.015 (25.78) | .745 (18.92) | 2.026 (51.46) | 2.281 (57.94) | .468 (11.89) | | .188 4/ (4.78) |
| 1 thru 8 (see note 8) | 66 | 2.420 (61.47) | 1.264 (32.10) | 1.115 (28.32) | 1.696 (43.08) | 1.969 (50.01) | .500 (12.70) | | .375 (9.52) |
| 1 thru 8 (see note 8) | 75 | 2.855 (72.52) | 1.370 (34.80) | 1.104 (28.04) | 2.026 (51.46) | 2.281 (57.94) | .234 (5.94) | .766 (19.46) | .203 (5.16) |
| 1 thru 8 (see note 8) | 104 | 2.900 (73.66) | 1.680 (42.67) | 1.476 (37.49) | 2.073 (52.65) | 2.375 (60.32) | .875 (22.22) | | .422 (10.72) |

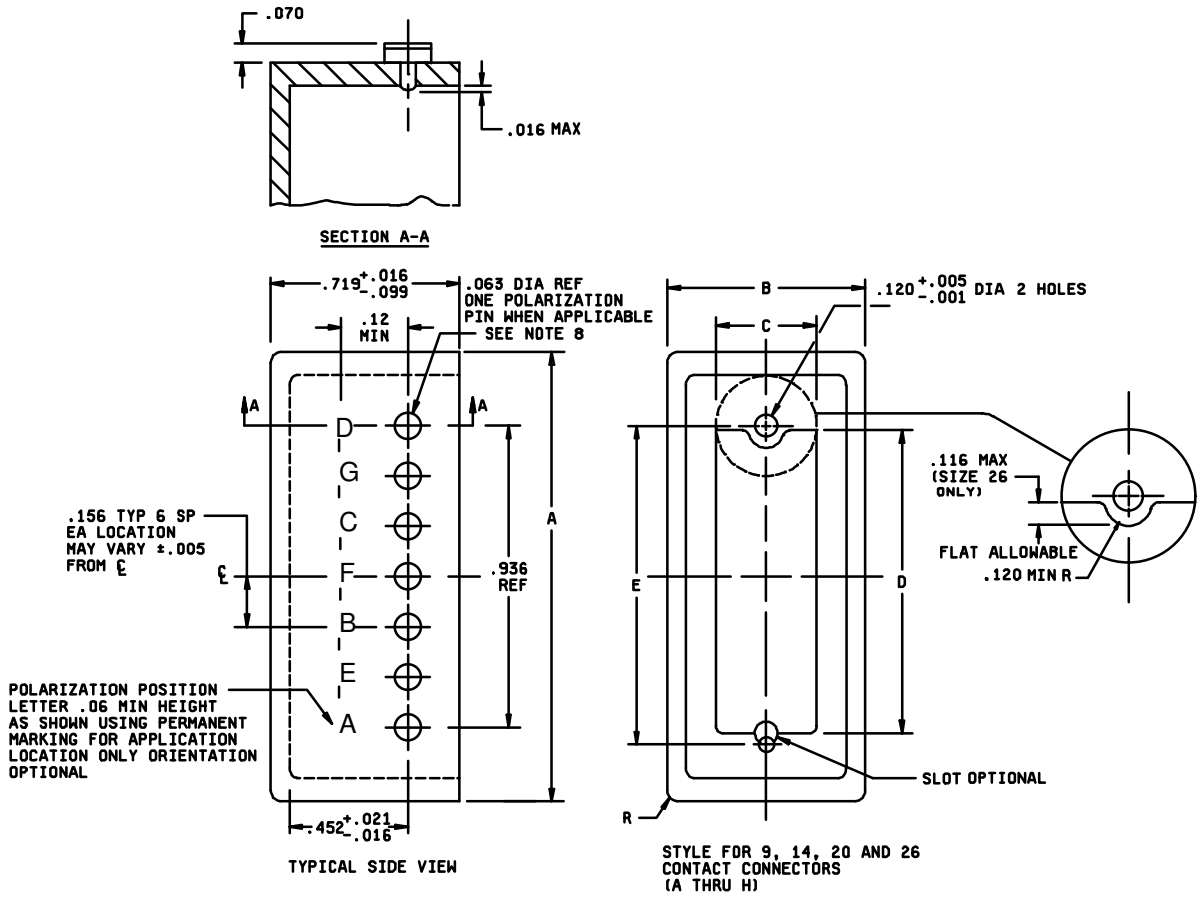
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Tolerance +.020 inch (0.51 mm) -.016 inch (0.41 mm).
5. Material aluminum alloy in accordance with specification QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
6. Finish anodize in accordance with MIL-A-8625, type I or type II.
7. Each slot may vary ±.005 inch (0.13 mm) from centerline.
8. Receptacle shell designators 1, 2, 3, 4, 5, 6, 7, and 8 are not used with one piece jackscrew shield designators 3, 4, 9, A, D, and E.

Receptacle shell polarizing

FIGURE 3. Shell configurations - Continued.



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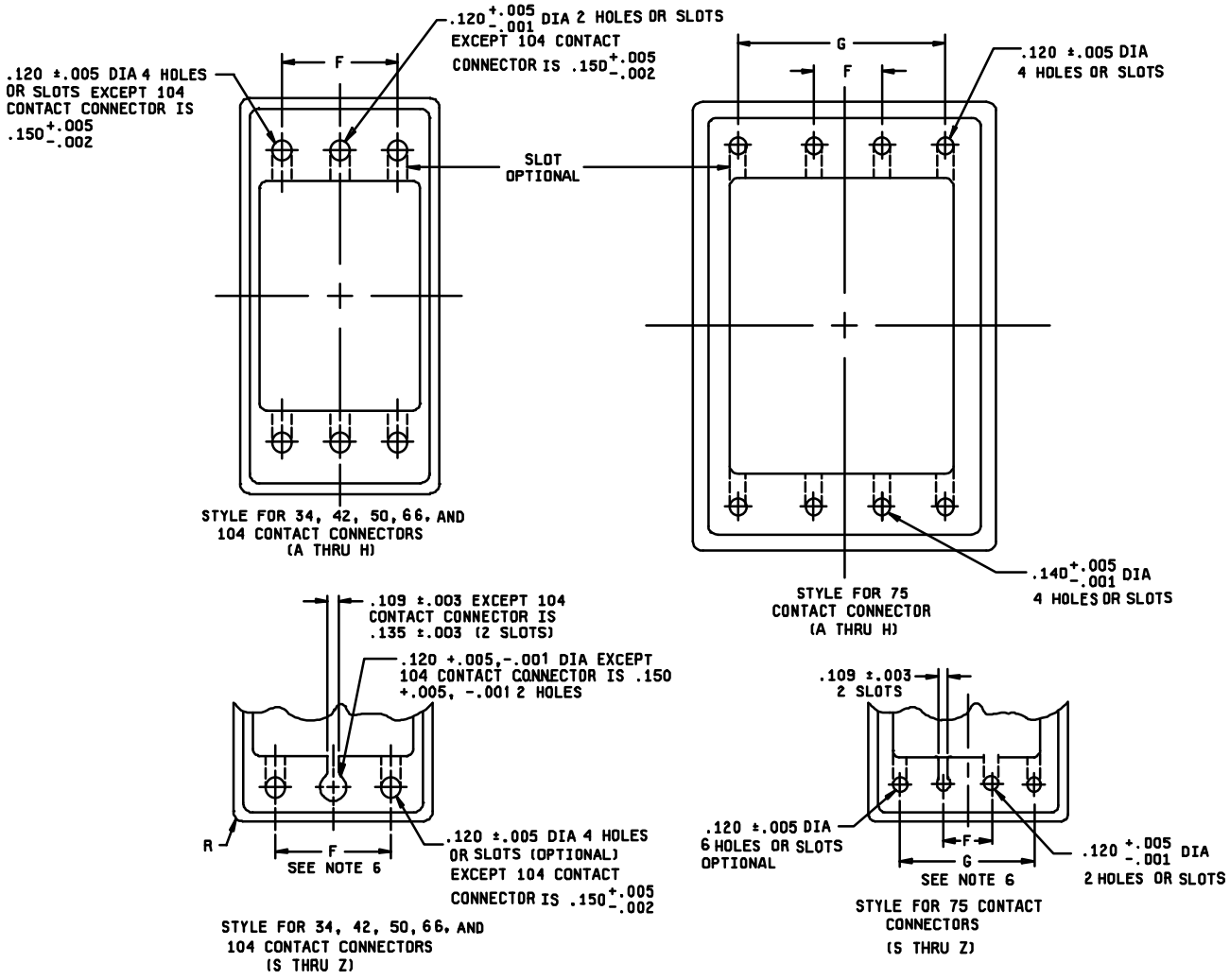



Plug shell polarizing

FIGURE 3. Shell configurations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .116 | 2.95 |
| .002 | 0.05 | .120 | 3.05 |
| .003 | 0.08 | .125 | 3.18 |
| .005 | 0.13 | .135 | 3.43 |
| .016 | 0.41 | .140 | 3.56 |
| .021 | 0.53 | .150 | 3.81 |
| .062 | 1.57 | .156 | 3.96 |
| .063 | 1.60 | .234 | 5.94 |
| .070 | 1.78 | .452 | 11.48 |
| .099 | 2.51 | .719 | 18.26 |
| .109 | 2.77 | .936 | 23.77 |

Plug shell polarizing

FIGURE 3. Shell configurations - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

| Shield and shield clamp location | Connector size (number of contacts) | A max | B max | C min | D min | E ±.005 (0.13 mm) | F ±.005 (0.13 mm) | G ±.005 (0.13 mm) | R |
|---|---|---|---|---|---|---|---|---|---|
| A thru H | 9 | 1.440 (36.58) | .500 (12.70) | .375 (9.52) | .864 (21.94) | 1.000 (25.40) | --- | | .141 (3.58) |
| A thru H | 14 | 1.380 (35.05) | .567 (14.40) | .440 (11.18) | .800 (20.32) | .938 (23.82) | --- | | .141 |
| A thru H | 20 | 1.700 (43.18) | .567 (14.40) | .440 (11.18) | 1.097 (27.86) | 1.250 (31.75) | --- | | .141 |
| A thru H | 26 | 1.755 (44.58) | .688 (17.48) | .565 (14.35) | 1.182 (30.02) | 1.312 (33.32) | --- | | .141 (3.58) |
| A thru H | 34 | 2.130 (54.10) | .880 (22.35) | .745 (18.92) | 1.400 (35.76) | 1.688 (42.88) | .468 (11.89) | | .141 (3.58) |
| A thru H | 42 | 2.442 (62.03) | .880 (22.35) | .745 (18.92) | 1.700 (43.18) | 2.000 (50.80) | .468 (11.89) | | .141 (3.58) |
| A thru H | 50 | 2.846 (72.39) | 1.005 (25.53) | .745 (18.92) | 2.026 (51.46) | 2.281 (57.94) | .468 (11.89) | | .188 (4.78) |
| A thru H | 66 | 2.410 (61.21) | 1.254 (31.85) | 1.115 (28.32) | 1.696 (43.08) | 1.969 (50.01 | .500 (12.70) | | .375 (9.52) |
| A thru H | 75 | 2.850 (72.39) | 1.365 (34.67) | 1.104 (28.04) | 2.026 (51.46) | 2.281 (57.94) | .234 (5.94) | .766 (19.46) | .187 (4.75) |
| A thru H | 104 | 2.890 (73.41) | 1.670 (42.42) | 1.476 (37.49) | 2.073 (52.65) | 2.375 (60.32) | .875 (22.22) | | .422 (10.72) |
| S thru Z (see note 6) | 34 | 2.130 (54.10) | .880 (22.35) | .745 (18.92) | 1.400 (35.56) | 1.688 (42.88) | .468 (11.89) | | .141 (3.58) |
| S thru Z (see note 6) | 42 | 2.442 (62.03) | .880 (22.35) | .745 (18.92) | 1.700 (43.18) | 2.000 (50.80) | .468 (11.89) | | .141 (3.58) |
| S thru Z (see note 6) | 50 | 2.846 (72.29) | 1.005 (25.53) | .745 (18.92) | 2.026 (51.46) | 2.281 (57.94) | .468 (11.89) | | .188 (4.78) |
| S thru Z (see note 6) | 66 | 2.410 (61.21) | 1.254 (31.85) | 1.115 (28.32) | 1.696 (43.08) | 1.969 (50.01) | .500 (12.70) | | .375 (9.52) |
| S thru Z (see note 6) | 75 | 2.850 (72.39) | 1.365 (34.67) | 1.104 (28.04) | 2.026 (51.46) | 2.281 (57.94) | .234 (5.94) | .766 (19.46) | .187 (4.75) |
| S thru Z (see note 6) | 104 | 2.890 (73.41) | 1.670 (42.42) | 1.476 (37.49) | 2.073 (52.65) | 2.375 (60.32) | .875 (22.22) | | .422 (10.72) |

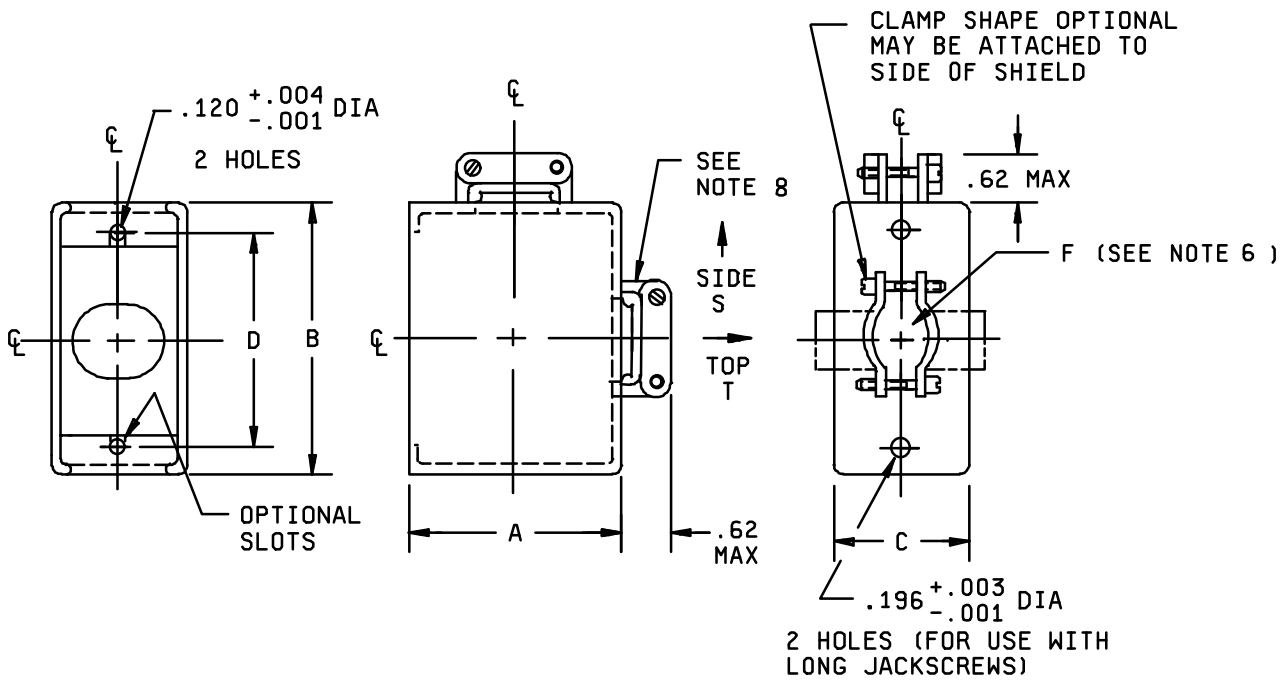
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodizes in accordance with MIL-A-8625, type I or type II.
6. Plug shell designators S, T, U, V, W, X, Y, and Z are not used with one piece jackscrew shield designators 9, A, D, and E.
7. One polarization pin is used when applicable.

Plug shell polarizing

FIGURE 3. Shell configurations - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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| Inches | mm |
|---|---|
| .001 | 0.03 |
| .003 | 0.08 |
| .004 | 0.10 |
| .120 | 3.05 |
| .196 | 4.98 |
| .62 | 15.75 |

Configurations 1 and 2 are used with shell for “L” hardware and can be used with or without shell for “M” hardware.
Configurations 3 and 4 are used without shell.

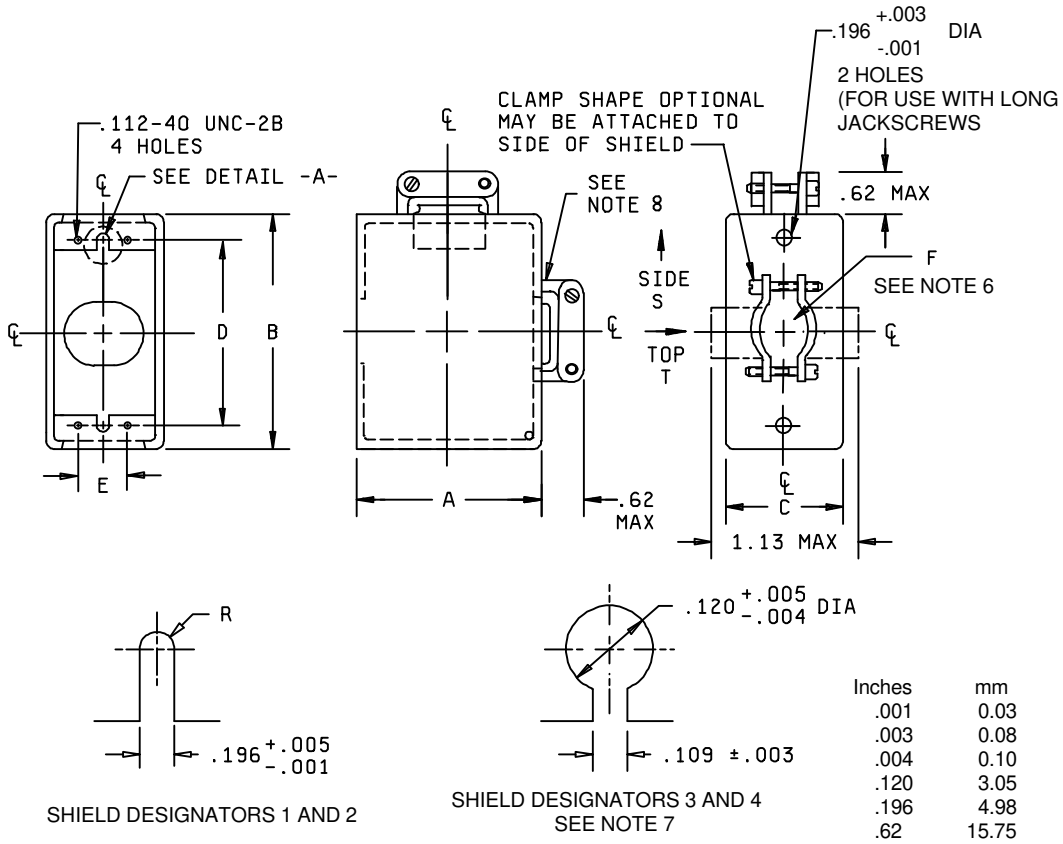
| Shield and shield clamp location | Connector size (no. of contacts) | A min max | B ± .016 (0.41 mm) | C ± .016 (0.41 mm) | D ±.005 (0.13 mm) | F Min. cable opening sq. inch |
|---|---|---|---|---|---|---|
| 1 - top opening<br>2 - side opening | 7 | 1.18 (29.97)<br>1.35 (34.29) | 1.220<br>(30.99) | .437<br>(11.10) | .906<br>(23.01) | .18<br>(4.57) |
| 1 - top opening<br>2 - side opening | 9 | 1.18 (29.97)<br>1.35 (34.29) | 1.312<br>(33.32) | .437<br>(11.10) | 1.000<br>(25.40) | .10<br>(2.54) |
| 1 - top opening<br>2 - side opening | 14 | 1.18 (29.97)<br>1.35 (34.29) | 1.250<br>(31.75) | .500<br>(12.70) | .938<br>(23.83) | .19<br>(4.83) |
| 1 - top opening<br>2 - side opening | 20 | 1.18 (29.97)<br>1.35 (34.29) | 1.562<br>(39.67) | .500<br>(12.70) | 1.250<br>(31.75) | .22<br>(5.59) |
| 1 - top opening<br>2 - side opening | 26 | 1.18 (29.97)<br>1.35 (34.29) | 1.625<br>(41.28) | .640<br>(16.26) | 1.312<br>(33.32) | .22<br>(5.59) |

See notes at end of figure.

FIGURE 4. Shield.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




Configurations 1 and 2 are used with shell for “L” hardware and can be used with or without shell for “M” hardware. Configurations 3 and 4 are used without shell.

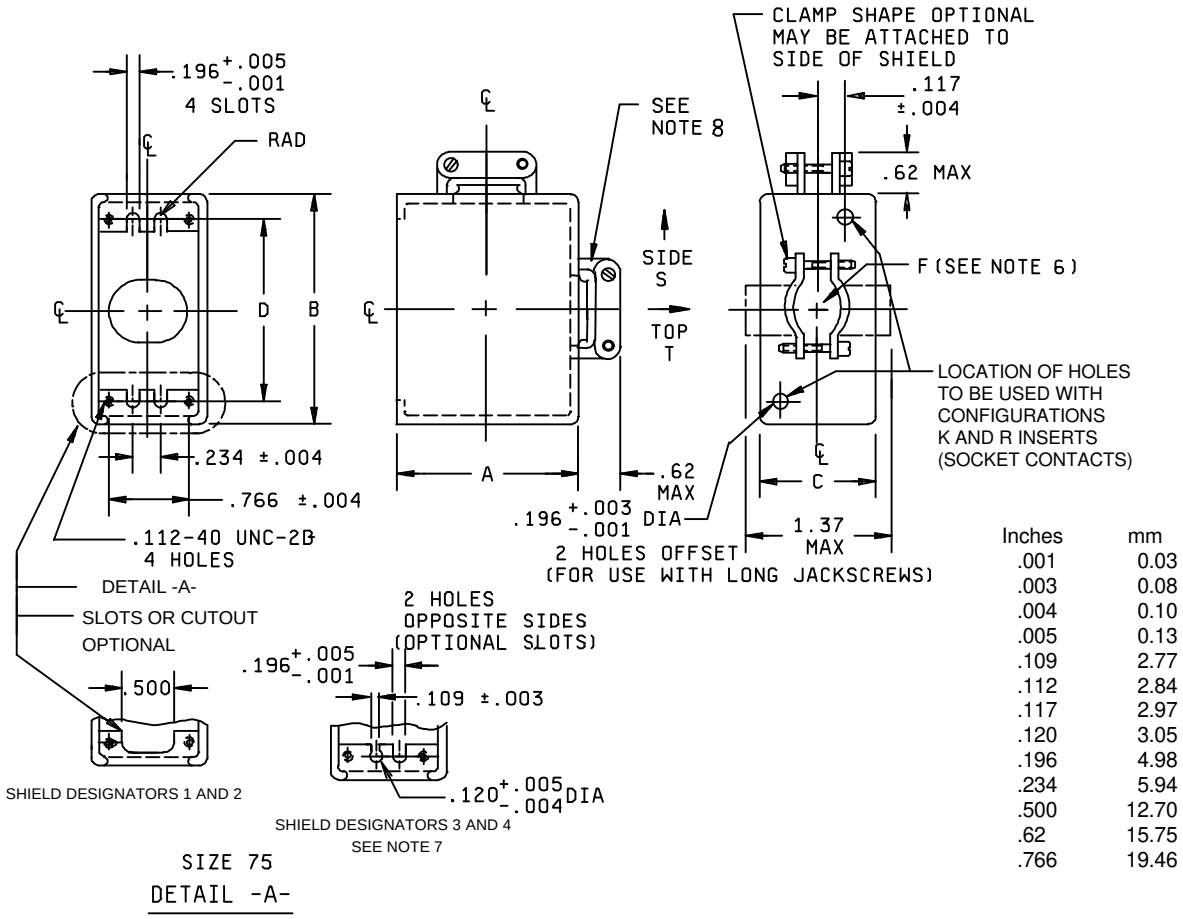
| Shield and shield clamp location | Connector size (no. of contacts) | A min max | B ± .016 (0.41 mm) | C ± .016 (0.41 mm) | D ±.005 (0.13 mm) | E ±.010 (0.25 mm) | F Min. cable opening sq. inch |
|---|---|---|---|---|---|---|---|
| 1 - top opening<br>2 - side opening | 34 | 1.22 (30.99)<br>1.35 (34.29) | 2.000<br>(50.80) | .828<br>(21.03) | 1.688<br>(42.88) | .468<br>(11.89) | .34<br>(8.64) |
| 1 - top opening<br>2 - side opening | 42 | 1.22 (30.99)<br>1.35 (34.29) | 2.312<br>(58.72) | .828<br>(21.03) | 2.000<br>(50.80) | .468<br>(11.89) | .34<br>(8.64) |
| 1 - top opening<br>2 - side opening | 50 | 1.22 (30.99)<br>1.35 (34.29) | 2.594<br>(65.89) | .828<br>(21.03) | 2.282<br>(57.96) | .468<br>(11.89) | .34<br>(8.64) |
| 3 - top opening<br>(see note 7) | 34 | 1.22 (30.99)<br>1.35 (34.29) | 2.000<br>(50.80) | .828<br>(21.03) | 1.688<br>(42.88) | .468<br>(11.89) | .34<br>(8.64) |
| 4 - side opening | | | | | | | |
| 3 - top opening<br>(see note 7) | 42 | 1.22 (30.99)<br>1.35 (34.29) | 2.312<br>(58.72) | .828<br>(21.03) | 2.000<br>(50.80) | .468<br>(11.89) | .34<br>(8.64) |
| 4 - side opening | | | | | | | |
| 3 - top opening<br>(see note 7) | 50 | 1.22 (30.99)<br>1.55 (39.37) | 2.594<br>(65.89) | .828<br>(21.03) | 2.282<br>(57.96) | .468<br>(11.89) | .34<br>(8.64) |
| 4 - side opening | | | | | | | |

See notes at end of figure.

FIGURE 4. Shield - Continued.



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| Inches | mm |
|---|---|
| .001 | 0.03 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .109 | 2.77 |
| .112 | 2.84 |
| .117 | 2.97 |
| .120 | 3.05 |
| .196 | 4.98 |
| .234 | 5.94 |
| .500 | 12.70 |
| .62 | 15.75 |
| .766 | 19.46 |

Configurations 1 and 2 are used with shell for “L” hardware and can be used with or without shell for “M” hardware.
Configurations 3 and 4 are used without shell.

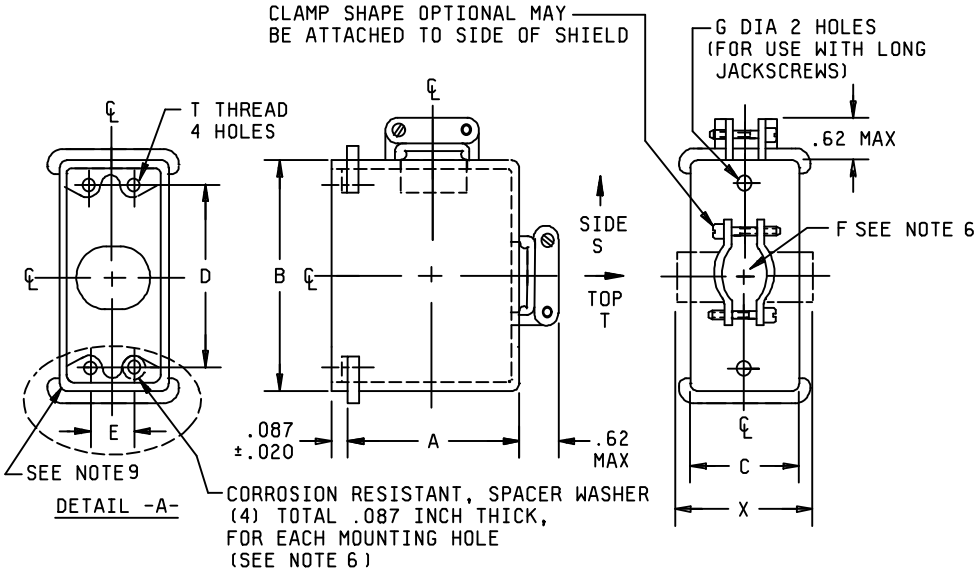
| Shield and shield clamp location | Connector size (no. of contacts) | A min max | B ± .016 (0.41 mm) | C ± .016 (0.41 mm) | D ±.005 (0.13 mm) | F Min. Cable opening sq. inch |
|---|---|---|---|---|---|---|
| 1 - top opening<br>2 - side opening | 75<br>Socket only | 1.22 (30.99)<br>1.35 (34.29) | 2.594<br>(65.89) | 1.188<br>(65.89) | 2.282<br>(57.96) | .55<br>(13.97) |
| 3 - top opening 9/<br>4 - side opening | 75<br>Socket only | 1.22 (30.99)<br>1.35 (34.29) | 2.594<br>(65.89) | 1.188<br>(65.89) | 2.282<br>(57.96) | .55<br>(13.97) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodizes in accordance with MIL-A-8625, type I or type II.
6. Edges of cable entry hole to be rounded to prevent insulation chafing.
7. Shield and shield clamp location part or identifying numbers (PIN) designators,” 3 and 4”, are for use with one-piece jackscrews.
8. Clamp hinge may attach to side of shield.

FIGURE 4. Shield - Continued.

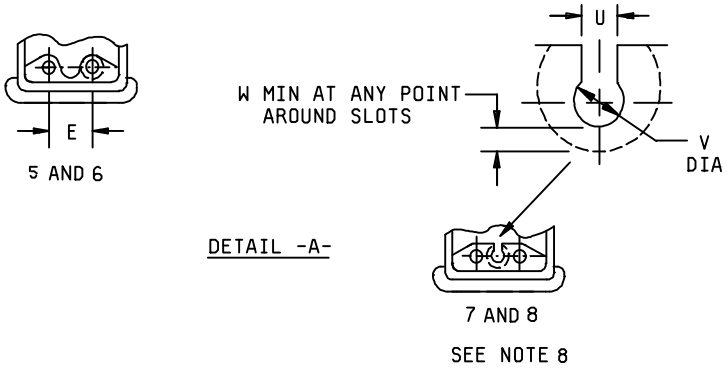


Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




| Shield and shield clamp location (see note 8) | U ±.003 (0.08 mm) | V +.005 (0.13 mm) -.001 (0.03 mm) | W Min | Connector size (no. of contacts) |
|---|---|---|---|---|
| 7 - top opening<br>8 - side opening | .109<br>(2.77) | .120<br>(3.05) | .062<br>(1.57) | 66 |
| 7 - top opening<br>8 - side opening | .133<br>(3.38) | .150<br>(3.81) | .110<br>(2.79) | 104 |

| Inches | mm |
|---|---|
| .001 | 0.03 |
| .003 | 0.08 |
| .004 | 0.10 |
| .020 | 0.51 |
| .087 | 2.21 |
| .120 | 3.05 |
| .196 | 4.98 |
| .62 | 15.75 |




Use with or without shells.

FIGURE 4. Shield - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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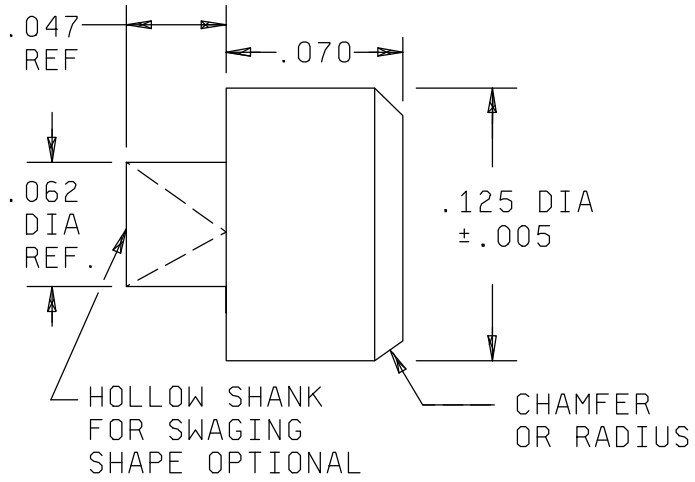
| Shield and shield clamp location | Connector size (no. of contacts) | A ±.03 (0.76 mm) | B ± .016 (0.41 mm) | C ± .016 (0.41 mm) | D ±.005 (0.13 mm) | E ±.005 (0.13 mm) | F Min. cable opening sq. inch | G +.003 (0.08 mm) -.001 (0.03 mm) | T thread | X max |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 - top opening<br>6 - side opening | 66 | 1.17<br>(29.72) | 2.366<br>(60.10) | 1.211<br>(30.76) | 1.969<br>(50.01) | .500<br>(12.70) | .55<br>(13.97) | .196<br>(4.98) | .112-40<br>UNC-2B | 1.37<br>(34.80) |
| 5 - top opening<br>6 - side opening | 104 | 2.23<br>(56.64) | 2.835<br>(72.01) | 1.625<br>(41.28) | 2.375<br>(60.33) | .875<br>(22.23) | 1.10<br>(27.94) | .229<br>(5.82) | .138-32<br>UNC-2B | 1.88<br>(47.75) |
| 7 - top opening<br>(see note 8)<br>8 - side opening | 66 | 1.17<br>(29.72) | 2.366<br>(60.10) | 1.211<br>(30.76) | 1.969<br>(50.01) | .500<br>(12.70) | .55<br>(13.97) | .196<br>(4.98) | .112-40<br>UNC-2B | 1.37<br>(34.80) |
| 7 - top opening<br>(see note 8)<br>8 - side opening | 104 | 2.23<br>(56.64) | 2.835<br>(72.01) | 1.625<br>(41.28) | 2.375<br>(60.33) | .875<br>(22.23) | 1.10<br>(27.94) | .229<br>(5.82) | .138-32<br>UNC-2B | 1.88<br>(47.75) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. Spacer washers, (4), used only with shell: discard in other uses.
7. Edges of cable entry hole to be rounded to prevent insulation chafing.
8. For use with one-piece jackscrews.
9. No cracks, chips, or sharp edges on all 4 edges of flange.

FIGURE 4. Shield - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .047 | 1.19 |
| .062 | 1.57 |
| .070 | 1.78 |
| .125 | 3.18 |

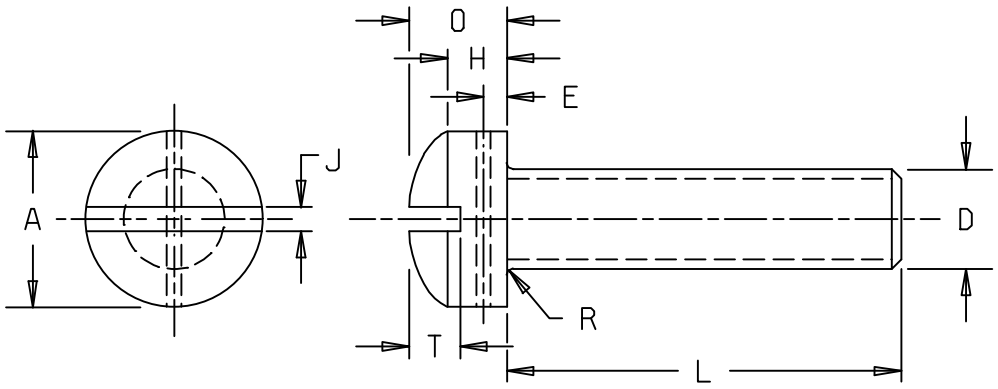
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.016 inch (0.41 mm).
3. Metric equivalents are given for general information only.
4. Material: Corrosion resistant steel in accordance with QQ-S-763, classes 302, 303, 303Se, 304, 316, or ASTM A484, or ASTM A532.
5. See M28748/1 to order polarizing pin separately.

Pin polarizing, straight headed.

FIGURE 5. Pin configuration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Screws furnished with shields when required, crossed drilled hole is optional.
To order screws individually use MS35275-215 (#4) or -229 (#6).

| Shield designator | Nominal size | Threads per inch, UNC-2A (Pitch) | D Body diameter | | A Head diameter | | H Head side height | | O Total head height | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | min | max | min | max | min | max | min | max |
| 1, 2, 3, 4 (sizes 34, 42, 50, 75) 5, 6, 7, 8 (size 66 only) | .112 (#4) (2.845) | 40 (.64) | .0925 (2.350) | .1120 (2.845) | .166 (4.216) | .183 (4.648) | .072 (1.829) | .079 (2.007) | .094 (2.388) | .107 (2.718) |
| 5, 6, 7, 8, Size 104 only | .138 (#6) (3.505) | 32 (.79) | .1141 (2.898) | .1380 (3.505) | .208 (5.283) | .226 (5.740) | .089 (2.261) | .096 (2.438) | .118 (2.997) | .132 (3.353) |

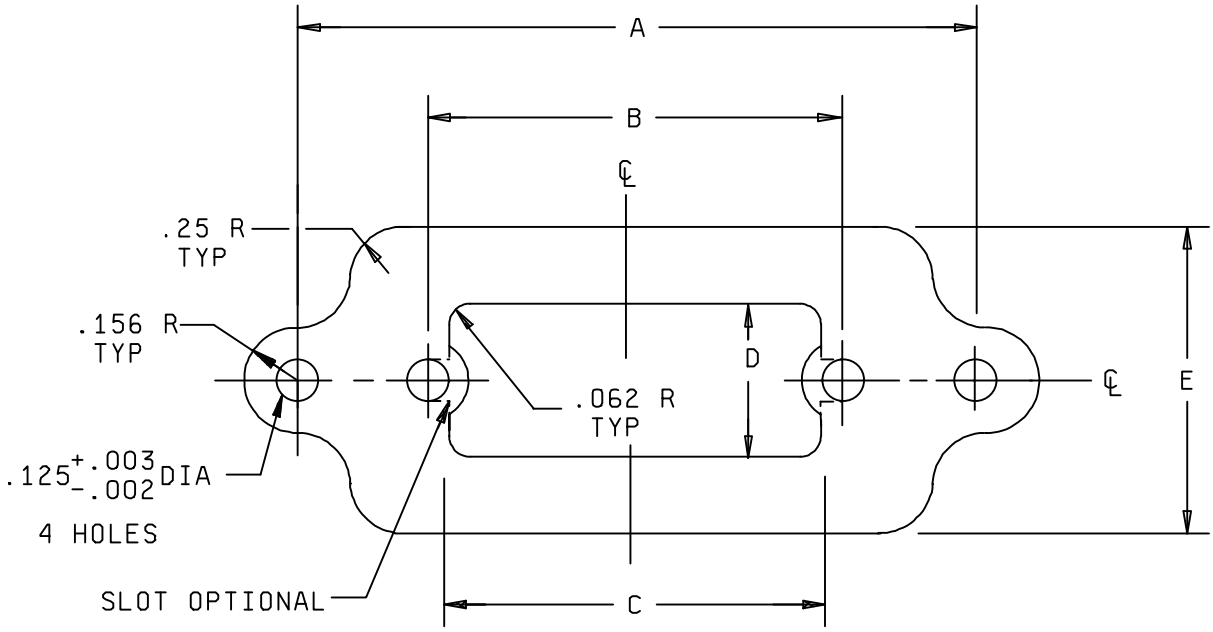
| Shield designator | J Slot width | | T Slot depth | | E drill location nom | Drilled hole size nom | R radius max | Tensile strength LB-min (KN-min) |
|---|---|---|---|---|---|---|---|---|
| | min | max | min | max | | | | |
| 1, 2, 3, 4 (sizes 34, 42, 50, 75) 5, 6, 7, 8 (size 66 only) | .031 (.787) | .039 (.991) | .030 (.762) | .038 (.965) | .035 (.889) | .037 (.940) | .017 (.432) | .480 (2.151) |
| 5, 6, 7, 8, Size 104 only | .039 (.991) | .048 (1.219) | .035 (.889) | .045 (1.143) | .043 (1.092) | .046 (1.168) | .021 (.533) | .730 (3.272) |

NOTES:
1. Material Corrosion resisting, in accordance with Federal standard number 66, AISI 302, 303, 304, 305, 316, 384 (developed for cold heading) or type UNS S30430 in accordance with the chemical requirements of ASTM-A-493.
2. Treatment: Passivation shall be in accordance with QQ-P-35, or an industry approved equivalent.
3. Magnetic permeability: These screws shall have a magnetic permeability or 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds, using a magnetic indicator in accordance with ASTM A342.
4. Threads: Threads shall be in accordance with FED-STD-H28.
5. Threads: Screws shall have complete threads extending to within two threads of the head, or closer if practicable.
6. Drilled hole is to be approximately at right angles to the slot and may be permitted to break through the bottom of the slot. All sharp edges shall be removed.

FIGURE 6. Slotted screw.



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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .062 | 1.57 |
| .125 | 3.18 |
| .156 | 3.96 |
| .25 | 6.35 |

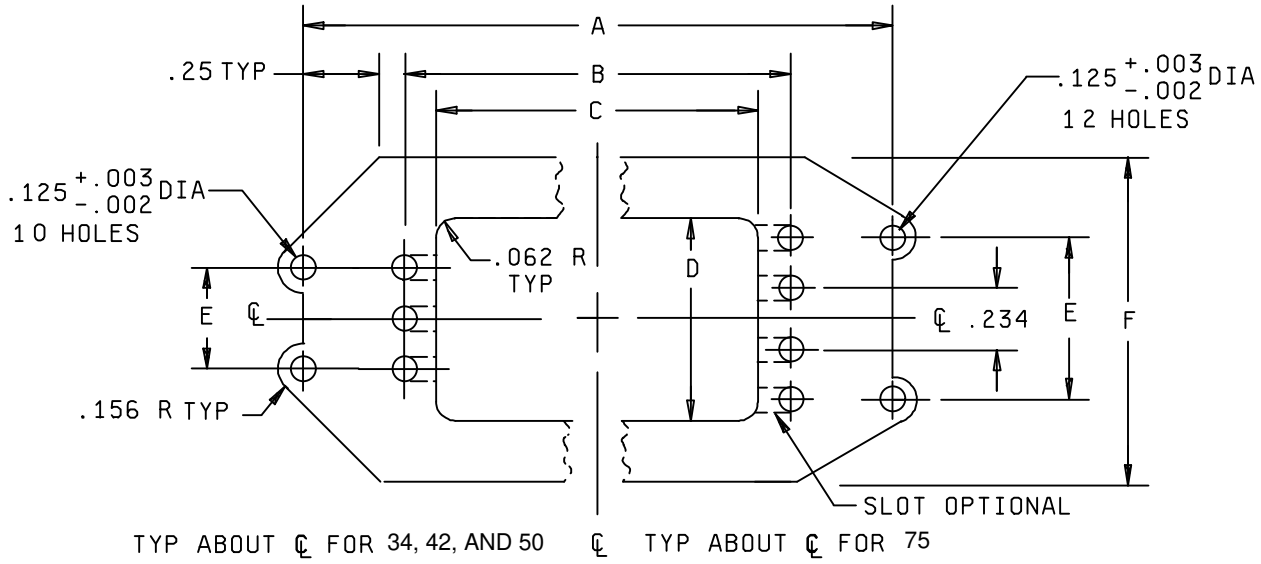
| Connector size (No. of contacts) | A ±.005 (0.13 mm) | B ±.005 (0.13 mm) | C | D | E |
|---|---|---|---|---|---|
| 9 | 2.016 (51.21) | 1.000 (2.54) | .875 (22.23) | .406 (10.31) | .953 (24.21) |
| 14 | 1.953 (49.61) | .937 (23.80) | .812 (20.62) | .469 (11.91) | 1.016 (25.81) |
| 20 | 2.266 (57.56) | 1.250 (31.75) | 1.125 (28.58) | .469 (11.91) | 1.016 (25.81) |
| 26 | 2.328 (59.13) | 1.312 (33.32) | 1.187 (30.15) | .593 (15.06) | 1.141 (28.98) |

See notes at end of figure.

FIGURE 7. Retaining plate.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .062 | 1.57 |
| .125 | 3.18 |
| .156 | 3.96 |
| .234 | 5.94 |
| .25 | 6.35 |

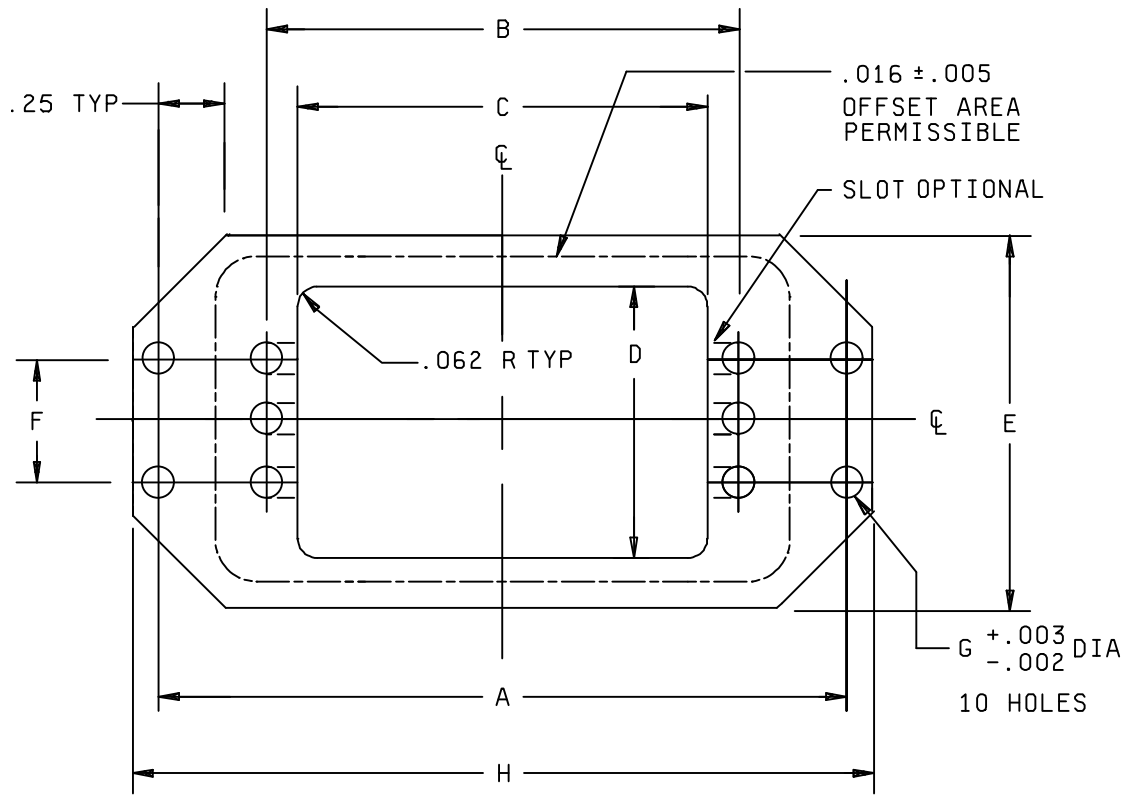
| Connector size (No. of contacts) | A ±.005 (0.13 mm) | B ±.005 (0.13 mm) | C | D | E ±.005 (0.13 mm) | F |
|---|---|---|---|---|---|---|
| 34 | 2.703 (68.66) | 1.687 (42.85) | 1.438 (36.53) | .750 (19.05) | .468 (11.89) | 1.328 (33.73) |
| 42 | 3.016 (78.89) | 2.000 (50.80) | 1.750 (44.45) | .750 (19.05) | .468 (11.89) | 1.328 (33.73) |
| 50 | 3.422 (86.92) | 2.281 (57.94) | 2.031 (51.60) | .750 (19.05) | .468 (11.89) | 1.453 (36.91) |
| 75 | 3.422 (86.92) | 2.281 (57.94) | 2.031 (51.60) | 1.109 (28.17) | 7.66 (19.46) | 1.797 (45.64) |

See notes at end of figure.

FIGURE 7. Retaining plate - Continued.



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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .016 | 0.41 |
| .062 | 1.57 |
| .25 | 6.35 |

| Connector size (No. of contacts) | A ±.005 (0.13 mm) | B ±.005 (0.13 mm) | C | D | E | F ±.005 (0.13 mm) | G | H |
|---|---|---|---|---|---|---|---|---|
| 66 | 2.984 (75.79) | 1.969 (50.01) | 1.718 (43.64) | 1.125 (28.58) | 1.688 (42.88) | .500 (17.70) | .125 (3.18) | 3.296 (83.72) |
| 104 | 3.500 (83.90) | 2.375 (60.33) | 2.125 (53.98) | 1.486 (37.74) | 2.000 (50.80) | .875 (22.23 | .150 (3.81) | 3.875 (98.43) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with specification QQ-A-250/8, temper H32, .040 inches (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.

FIGURE 7. Retaining plate - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, 3, 4, and 5.

Mating connector: See MIL-DTL-28748/1 and MIL-DTL-28748/3.

Part or Identifying Number (PIN): As shown in the following example: 1/

(See (PIN) code)

| M28748/4 | B | A | 1 | H | L | 1 A |
|---|---|---|---|---|---|---|
| Basic number of specification sheet | Hardware designator (see figure 1 for configu-ration styles) | Insert designator (see figure 2) | Shield and shield clamp location or retaining plate (see figure 4) | Shell and polarization position (see figure 3) | Jackscrews or guidepins (see figure 1) | Contacts |

PIN code:

| Hardware 2/ designators | Insert designators | Number of contact positions |
|---|---|---|
| A - Long jackscrew hardware only. | A | 7 |
| B - All other hardware. | B | 9 |
| - - (insert only) | C | 14 |
| | D | 20 |
| | E | 26 |
| | F | 34 |
| | G | 42 |
| | H | 50 |
| | J | 66 |
| | K | 75 |
| | L | 104 |
| | 0 | No insert included |
| | Insert designators through hole 4/ | |
| | M | 34 |
| | N | 42 |
| | P | 50 |
| | Q | 66 |
| | R | 75 |
| | S | 104 |
| | Insert designators for piece part orders 10/ | Connector sizes |
| | T | 7 |
| | U | 9 |
| | V | 14 |
| | W | 20 |
| | X | 26 |
| | Y | 34 |
| | Z | 42 |
| | 1 | 50 |
| | 2 | 66 |
| | 3 | 75 |
| | 4 | 104 |

| Shield and shield clamp Locations | Polarization designators | Polarization positions 3/ |
|---|---|---|
| | 0 - None included | |
| 1 - top opening 5/ | A | A |
| 2-side opening 5/ | B | B |
| 3 - top opening 6/ | C Plug | C |
| 4-side opening 6/ | D shells -- | D |
| 5 - top opening 7/ | E | E |
| 6-side opening 7/ | F | F |
| 7- top opening 7/ | G | G |
| 8- side opening 7/ | H - Unpolarized | |
| | J | A |
| | K | B |
| 0-none included | L Receptacle | C |
| | M shells ---------- | D |
| | N | E |
| Retaining plate | P | F |
| P - Plate | Q | G |
| | R - Unpolarized | |
| | S | A |
| | T | B |
| | U Plug | C |
| | V shells -- | D |
| | W | E |
| | X | F |
| Through hole insert designators for piece part orders 9/ | Y | G |
| | Z - Unpolarized | |
| --- | 1 | A |
| --- | 2 | B |
| --- | 3 Receptacle | C |
| --- | 4 shells ---------- | D |
| 5 | 5 | E |
| 6 | 6 | F |
| 7 | 7 | G |
| 8 | 8 - Unpolarized | |
| 9 | | |
| A | | |

See footnotes on next page



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Jackscrews
or
guidepins

F - Fixed jackscrews, configuration C, figure 1 (not used with shield).
G - Guidepins, configuration B, figure 1.
L - Long jackscrews, one-piece construction, configuration E, figure 1.
M - Long jackscrews, multi-piece construction, configuration F, figure 1.
S - Short jackscrews slotted head, configuration D, figure 1 (not used with shield).
0 - None included.

Jack and guide location: The jackscrew male (turning or fixed) or guide pin shall be located by the A or 1 pin contact.

Contacts: In accordance with MIL-DTL-28748.

1A - 100 percent size 16
1L - none included 10/

1/ The number zero (0) is used to indicate which parts are not included.
2/ Used for information to indicate size and style of insert or hardware to be used.
3/ Polarization positions are for location of plug and receptacle shells pins and slots (see figure 3).
4/ Manufacturers, mold bodies, and hardware will not necessarily interchange with each other.
5/ Used when shell is specified and with “L” hardware or used with or without shell for “M” hardware.
6/ Use when no shell is specified and with “L” hardware.
7/ Used with or without shell for “M” hardware or used when shell is specified and with “L” hardware.
8/ Used when no shell is specified and with “L” hardware.
9/ These designators are informational and used for insert reference (connector size) to order an appropriate shield or shell as a piece part. Shields and shells may be ordered together or separately.
10/ The “1L” ordering code shall be used in conjunction with insert designator zero “0” or with the insert designators for piece orders for which no insert is included.



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TABLE I. Contact size percent.

| Contact | Sizes 16-16 M39029/35-276 (percent) | Sizes 16-20 M39029/35-275 (percent) | Sizes 20-20 M39029/35-274 (percent) |
|---|---|---|---|
| 1A | 100 | 0 | 0 |
| 1B | 90 | 0 | 0 |
| 1C | 80 | 0 | 0 |
| 1D | 70 | 0 | 0 |
| 1E | 60 | 0 | 0 |
| 1F | 50 | 0 | 0 |
| 1G | 40 | 0 | 0 |
| 1H | 30 | 0 | 0 |
| 1J | 20 | 0 | 0 |
| 1K | 10 | 0 | 0 |
| 1L | 0 | 0 | 0 |
| 2A | 0 | 100 | 0 |
| 2B | 0 | 90 | 0 |
| 2C | 0 | 80 | 0 |
| 2D | 0 | 70 | 0 |
| 2E | 0 | 60 | 0 |
| 2F | 0 | 50 | 0 |
| 2G | 0 | 40 | 0 |
| 2H | 0 | 30 | 0 |
| 2J | 0 | 20 | 0 |
| 2K | 0 | 10 | 0 |
| 3A | 0 | 0 | 100 |
| 3B | 0 | 0 | 90 |
| 3C | 0 | 0 | 80 |
| 3D | 0 | 0 | 70 |
| 3E | 0 | 0 | 60 |
| 3F | 0 | 0 | 50 |
| 3G | 0 | 0 | 40 |
| 3H | 0 | 0 | 30 |
| 3J | 0 | 0 | 20 |
| 3K | 0 | 0 | 10 |

Crimp tool M22520/1-01
Positioner M22520/1-03(A)
Installing tool M81969/18-01.
Removal tool M81969/20-01.



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TABLE II. Superseded specification and MS sheets.

| Specification or MS sheet | Dated |
|---|---|
| MIL-C-28748/4D | 18 March 1985 |
| MS14005 | 18 March 1985 |
| MS14006 | 18 March 1985 |
| MS14011 | 18 March 1985 |
| MS18173 | 20 May 1970 |
| MS18175 | 2 August 1982 |
| MS18177 | 18 March 1985 |
| MS18179 | 9 February 1987 |
| MS18181 | 27 November 1985 |
| MS18183 | 18 March 1985 |
| MS18185 | 18 March 1985 |
| MS18188 | 18 July 1988 |
| MS18190 | 18 March 1985 |
| MS18192 | 1 June 1983 |
| MS18193 | 26 April 1990 |
| MS18198 | 23 June 1972 |
| MS18199 | 23 June 1972 |
| MS18200 | 20 May 1970 |
| MS18203 | 26 April 1990 |
| MS18204 | 26 April 1990 |
| MS18205 | 23 June 1972 |
| MS35275 | 14 June 1976 |
| MS35275 | 14 June 1976 |

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 17, 99

Preparing activity:
DLA - CC

(Project 5935-4132-04)



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
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| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER** MIL-DTL-28748/4E | **2. DOCUMENT DATE** *(YYYYMMDD)* 1999 May 11 |
|---|---|---|

**3. DOCUMENT TITLE**
CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, CRIMP TYPE REMOVABLE SOCKET CONTACTS, SIZES 16 AND 20

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include zip code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) DSN *(if applicable)* | 7. DATE SUBMITTED *(YYYYMMDD)* |

**8. PREPARING ACTIVITY**

| a. NAME<br>**Defense Logistics Agency**<br>**Defense Supply Center, Columbus** | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial **614-692-0538**<br>(2) DSN **850-0538** |
|---|---|
| c. ADDRESS *(Include Zip Code)*<br>**DSCC-VAI**<br>**P.O. Box 3990**<br>**Columbus, Ohio 43216-5000** | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>**Defense Standardization Program Office (DLSC-LM)**<br>**8725 John J. Kingman Road,Suite 2533**<br>**Fort Belvoir, Virginia 22060-6621**<br>**Telephone (703) 767-6888 DSN 427-6888** |

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